AISSMS
INSTITUTE OF INFORMATION TECHNOLOGY
ADDING VALUE TO ENGINEERING

DEPARTMENT OF COMPUTER ENGINEERING

Third Year of Computer Engineering

(2022 Course)

Laboratory Manual

for

A.Y. 2025-2026

COVSE606:

Robotic Process Automation

Laboratory

List of Experiments

SUBJECT**: Robotic Process Automation Laboratory**

CLASS: T.Y. SEMESTER: VI

Academic Year :

Sr. No.	Title of Experiment	CO	PO	PSO
1.	UiPath Studio/Orchestrator Installation	C606.1	1,3,5	1,3
2.	Build a simple calculator with operations 1.Add 2. Subtract 3.Multiplication 4. Division	C606.2	1,3,5	1,3
3.	Use of control flow using if-else statement and Switch statement.	C606.3	1,3,5	1,3
4.	Excel file automation	C606.3	1,3,5	1,3
5.	Email automation	C606.3	1,3,5	1,3
6.	Keyboard, mouse automation	C606.3	1,3,5	1,3
7.	Generate sharable workflow in Uipath	C606.4	1,3,5	1,3
8.	In a text file find a word and replace it with another word	C606.5	1,3,5	1,3
9.	Read text file and write the contents in another file. Extract contents of text file in message box.	C606.5	1,3,5	1,3
10.	Web browser automation	C606.6	1,3,5	1,3
11.	Web scraping automation	C606.6	1,3,5	1,3
12.	PDF automation	C606.6	1,3,5	1,3
13.	Mini-project	C606	1,3,5	1,3

Mrs. Supriya Deshmukh / Mrs. Bhavana Zambare **Dr. K. S. Wagh**

Subject In-charge **H.O.D**

Experiment 1

Title: UiPath Studio Installation

Aim: To successfully install UiPath Studio on a Windows system, enabling the development and automation of tasks using Robotic Process Automation (RPA) tools.

Objectives:

1. To download the UiPath Studio installer from the official website.
2. To set up a UiPath account and sign in to the Automation Cloud.
3. To install UiPath Studio and configure the environment.
4. To verify the installation by launching UiPath Studio and creating a sample project.

Theory:

UiPath Studio is a leading Robotic Process Automation (RPA) development environment that enables individuals and organizations to automate repetitive tasks in a highly visual and efficient manner. RPA, or Robotic Process Automation, is a technology that allows software robots (also called "bots") to mimic human actions on a computer, such as clicking, typing, copying data, and interacting with websites or applications.

Introduction to UiPath Studio

UiPath Studio provides a user-friendly interface where developers and even non-programmers can design automation workflows using a drag-and-drop approach. It supports both **low-code** and **no-code** automation, making it accessible to beginners as well as powerful enough for advanced developers. With UiPath, automation workflows are built using **activities**, which are pre-defined functions such as reading from Excel, sending an email, scraping data from a website, or interacting with files and applications.

The tool uses **Visual Basic.NET (VB.NET)** under the hood but hides much of the complexity through its intuitive interface. It supports a variety of automation scenarios, including front-office automation (interacting with desktop applications and browsers) and back-office automation (handling background tasks like file and database operations).

Components of UiPath Platform

The UiPath ecosystem includes several core components:

- **UiPath Studio**: The primary development environment for building automation workflows.
- **UiPath Robot**: Executes the automation created in UiPath Studio.

- **UiPath Orchestrator**: A web-based platform that manages, schedules, and monitors automated tasks across multiple machines.
- **UiPath Assistant**: A desktop application that allows users to trigger attended bots.

Types of Installation

There are two major editions of UiPath Studio:

1. **Community Edition (Free)** – Ideal for students, individual developers, and small teams.
2. **Enterprise Edition (Paid)** – Comes with premium features, support, and scalability options suited for business use.

The Community Edition is the most commonly used version for learning and training purposes. It provides almost all the core features needed to build, test, and run RPA workflows.

System Requirements

Before installation, it is important to ensure your system meets the following prerequisites:

- **Operating System**: Windows 10 or later (64-bit)
- **RAM**: Minimum 4 GB (8 GB recommended)
- **.NET Framework**: Version 4.7.2 or later
- **Internet Access**: Required for activation and downloading updates
- **UiPath Account**: Required for signing into Automation Cloud

Installation Process Overview

1. **Sign Up on UiPath Automation Cloud**
   Users must first create an account at https://cloud.uipath.com. This cloud platform allows access to Studio, Orchestrator, and other services.
2. **Download Installer**
   After logging into the Automation Cloud, users can download the latest version of the **UiPath Studio Installer (.exe)**.
3. **Run Installer and Choose Profile**
   During the installation process, the user selects the type of profile (e.g., UiPath Studio, UiPath StudioX for business users). The installer downloads necessary components and completes the setup.
4. **Sign In to Activate**
   Once installed, the user signs in with their UiPath account to activate the Community Edition license.

5. **Create a New Project**
   After installation, users can create their first project and test the environment using simple activities like **Message Box**, **Assign**, or **Write Line**.

Post-Installation: Verifying Setup

To verify the setup, a basic automation such as displaying a "Hello World" message box can be created. This serves as a functional check to ensure that the UI is working, the robot can execute workflows, and activities can be added and executed without errors.

Benefits of UiPath Studio

- **Ease of Use**: Drag-and-drop UI, low-code environment.
- **Integration**: Works with Excel, Word, PDF, email, APIs, and databases.
- **Extensibility**: Supports custom libraries and reusable components.
- **Debugging and Logging**: Built-in tools to test and monitor workflows.
- **Scalability**: Can scale from personal automation to enterprise-level deployments.

Program Input and Output :

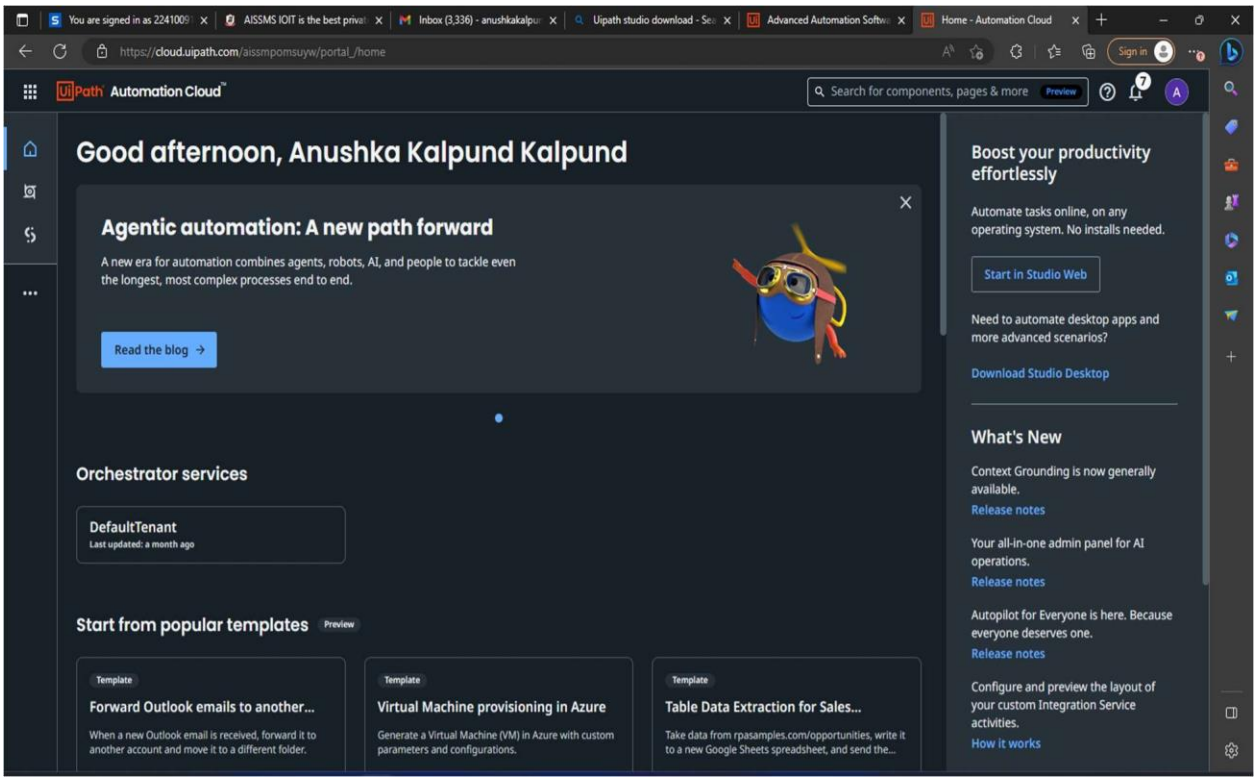
Input:

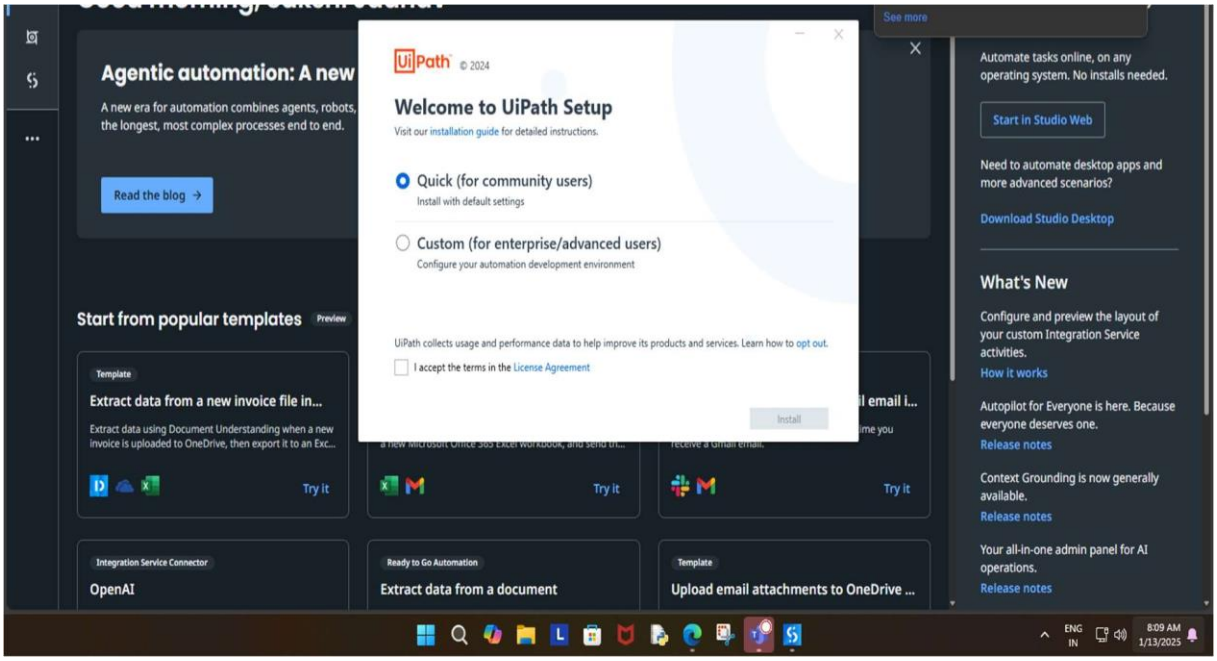
No user input required for this sample program.

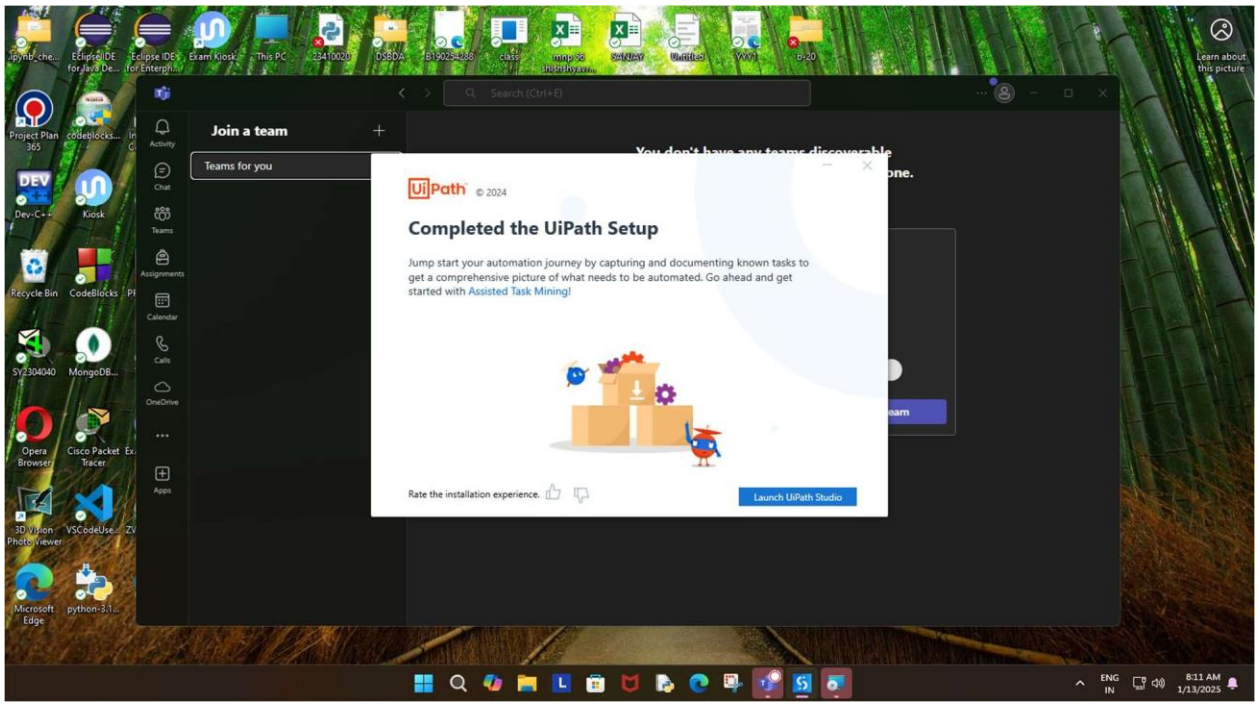
Output: (on next page)

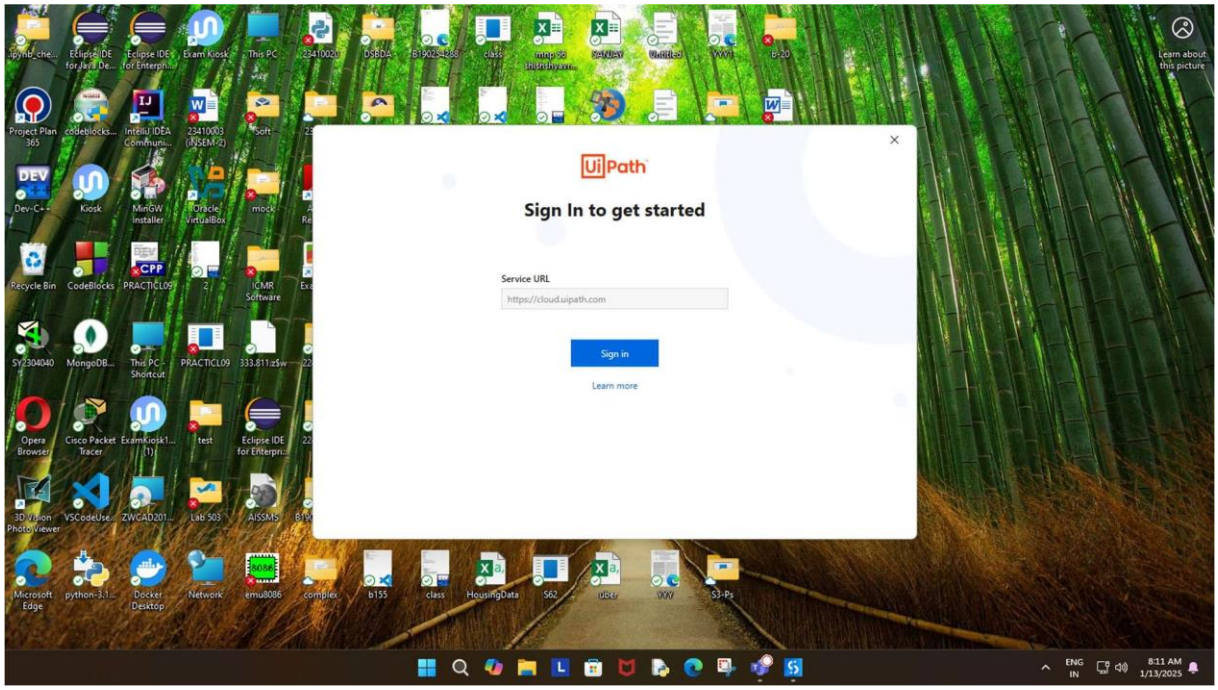
Conclusion :

UiPath Studio was successfully installed and verified through a sample automation. The setup provides a ready-to-use environment for developing RPA workflows. With this installation, users can begin building automation projects to enhance productivity and eliminate manual, repetitive tasks.


Home - Automation Cloud
https://cloud.uipath.com/aissmpomsuyw/portal_/home
UiPath Automation Cloud
Search for components, pages & more
Good afternoon, Anushka Kalpund Kalpund
Agentic automation: A new path forward
A new era for automation combines agents, robots, AI, and people to tackle even the longest, most complex processes end to end.
Read the blog
Orchestrator services
DefaultTenant
Last updated: a month ago
Start from popular templates
Preview
Template
Forward Outlook emails to another...
When a new Outlook email is received, forward it to another account and move it to a different folder.
Template
Virtual Machine provisioning in Azure
Generate a Virtual Machine (VM) in Azure with custom parameters and configurations.
Template
Table Data Extraction for Sales...
Take data from rpasamples.com/opportunities, write it to a new Google Sheets spreadsheet, and send the...
Boost your productivity effortlessly
Automate tasks online, on any operating system. No installs needed.
Start in Studio Web
Need to automate desktop apps and more advanced scenarios?
Download Studio Desktop
What's New
Context Grounding is now generally available.
Release notes
Your all-in-one admin panel for AI operations.
Release notes
Autopilot for Everyone is here. Because everyone deserves one.
Release notes
Configure and preview the layout of your custom Integration Service activities.
How it works


Agentic automation: A new
A new era for automation combines agents, robots,
the longest, most complex processes end to end.
Read the blog
UiPath © 2024
Welcome to UiPath Setup
Visit our installation guide for detailed instructions.
Quick (for community users)
Install with default settings
Custom (for enterprise/advanced users)
Configure your automation development environment
UiPath collects usage and performance data to help improve its products and services. Learn how to opt out.
I accept the terms in the License Agreement
Install
Start from popular templates
Preview
Template
Extract data from a new invoice file in...
Extract data using Document Understanding when a new invoice is uploaded to OneDrive, then export it to an Exc...
Try it
Try it
Try it
Integration Service Connector
OpenAI
Ready to Go Automation
Extract data from a document
Template
Upload email attachments to OneDrive ...
See more
Automate tasks online, on any operating system. No installs needed.
Start in Studio Web
Need to automate desktop apps and more advanced scenarios?
Download Studio Desktop
What's New
Configure and preview the layout of your custom Integration Service activities.
How it works
Autopilot for Everyone is here. Because everyone deserves one.
Release notes
Context Grounding is now generally available.
Release notes
Your all-in-one admin panel for AI operations.
Release notes
ENG IN
8:09 AM 1/13/2025


Join a team
Teams for you
UiPath © 2024
Completed the UiPath Setup
Jump start your automation journey by capturing and documenting known tasks to get a comprehensive picture of what needs to be automated. Go ahead and get started with Assisted Task Mining!
Rate the installation experience.
Launch UiPath Studio


UiPath
Sign In to get started
Service URL
https://cloud.uipath.com
Sign in
Learn more

Experiment 2

Title: Build a simple calculator with opeartion 1.Add 2.Subtract 3.Multiplication 4.Division

Aim: To develop a simple calculator in UiPath Studio that performs basic arithmetic operations — Addition, Subtraction, Multiplication, and Division — using user input and decision-making activities.

Objectives:

1. To design a workflow in UiPath Studio that accepts two numerical inputs from the user.
2. To provide a choice-based selection for performing arithmetic operations.
3. To implement conditional logic (e.g., If, Switch) for selecting and executing the chosen operation.
4. To display the result using message boxes or write line activities.
5. To understand the application of variables, expressions, and control structures in UiPath.

Theory:

UiPath Studio is a leading Robotic Process Automation (RPA) platform that enables users to automate tasks through visual workflows. It supports both **attended** and **unattended** automation and is widely used to simulate human actions in computer-based tasks.

One of the key strengths of UiPath is its **low-code development environment**, where workflows are designed using pre-defined activities such as Assign, Input Dialog, If, Switch, Message Box, etc.

In this project, we are building a **basic calculator** using UiPath Studio. The calculator accepts two numbers and a user's choice of operation (Add, Subtract, Multiply, Divide), then performs the appropriate computation and displays the result.

To achieve this, the following core concepts of UiPath are used:

Variables:

Variables are used to store and manipulate values throughout the automation. Here, we use:

- num1, num2: For storing input numbers.
- choice: For storing the user's operation selection.

- result: For storing the computed result.

Input Dialog:

Used to get user input during execution. It can be used for entering numbers and operation choices.

Switch Activity:

The Switch activity in UiPath allows multiple branching paths based on a single condition. It's ideal for handling menu-based selection like calculator operations.

Data Conversion:

Since all inputs via Input Dialog are **strings**, we convert them to **Double** or **Integer** using Convert.ToDouble(string) or Convert.ToInt32(string) before performing arithmetic operations.

Message Box:

Used to display the result to the user in a popup window.

Error Handling (optional):

Though not mandatory in basic calculators, error handling mechanisms like Try Catch can be added to manage invalid inputs, especially for division by zero.

This project is a good demonstration of how UiPath can be used not just for enterprise automation but also for logical and interactive applications — making it a great learning exercise for beginners.

Program Input and Output :

Input:

Enter first number: 20

Enter second number: 30

Choose operation:

1. Addition

2. Subtraction

3. Multiplication

4. Division

User selects: 1

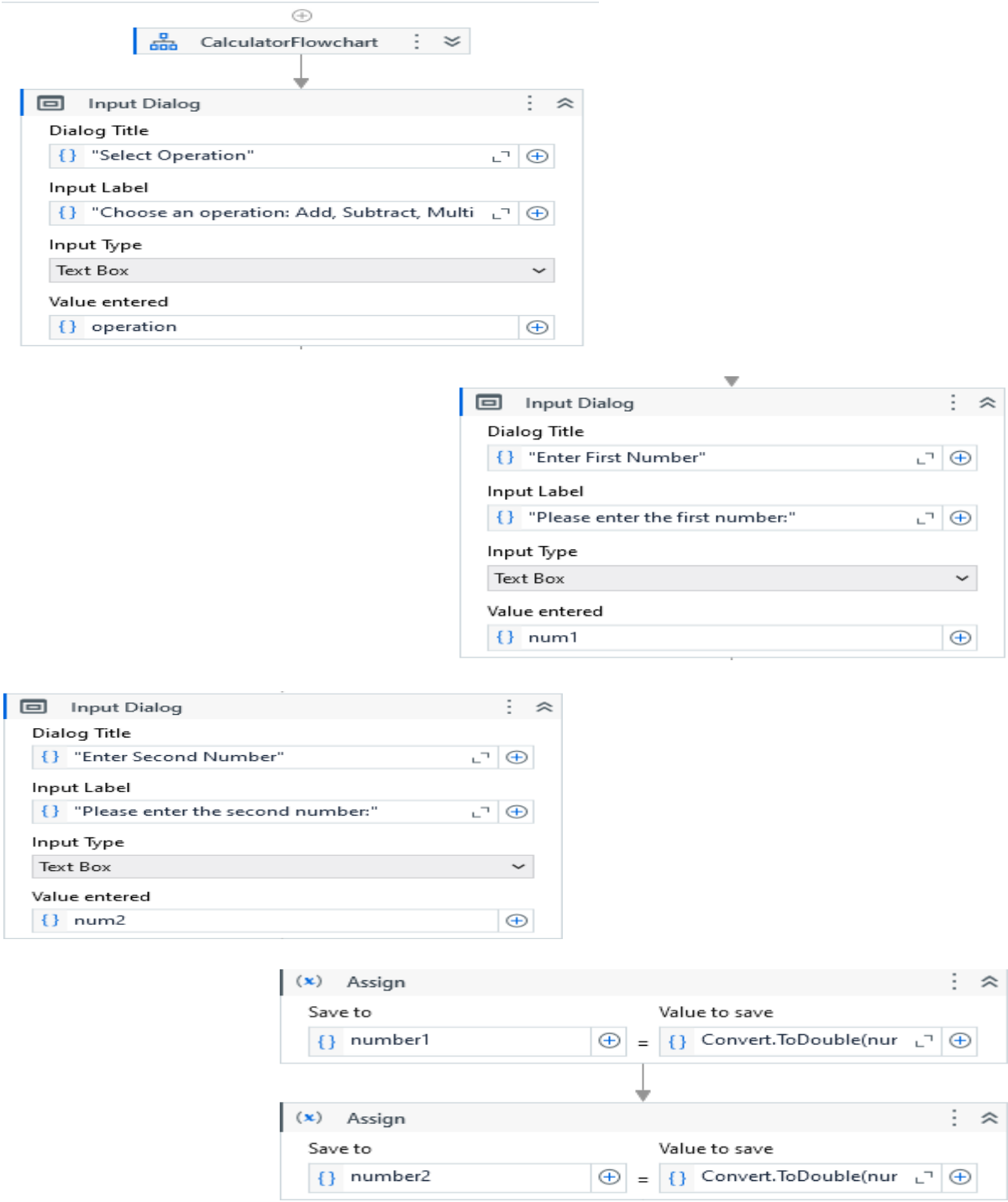
Output:

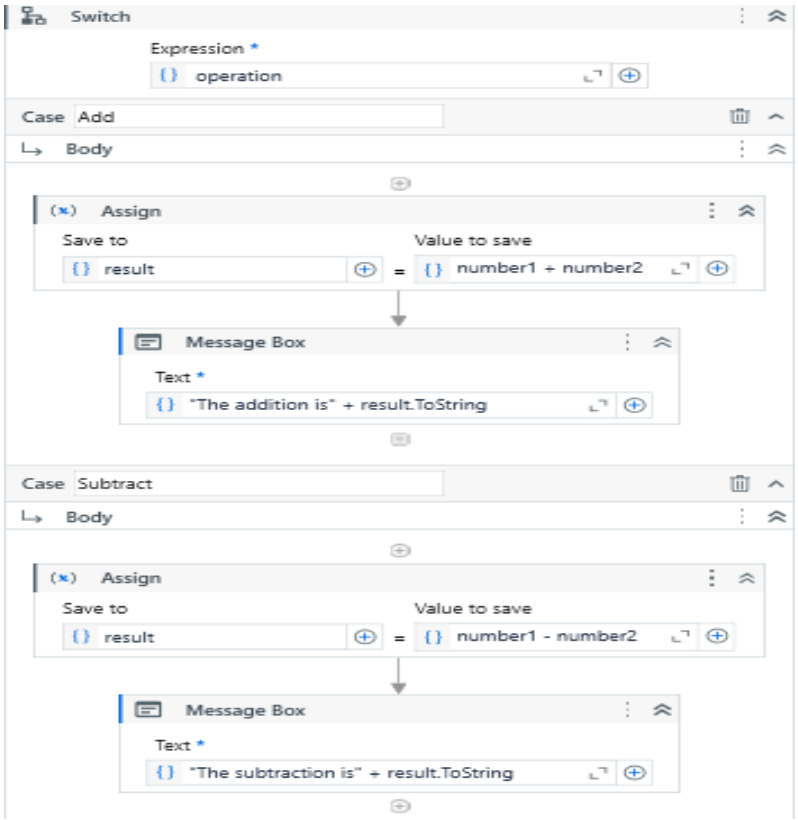
Result: 50

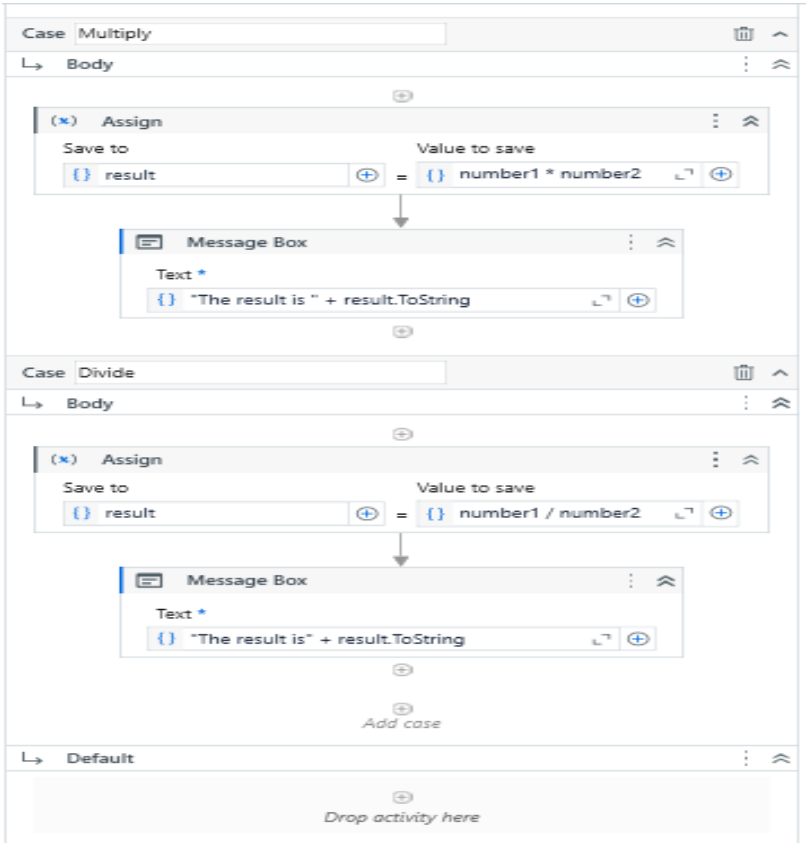
Conclusion :

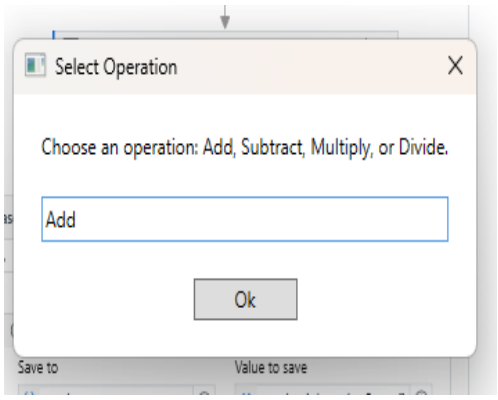
The simple calculator was successfully implemented using UiPath Studio. This project demonstrated the use of variables, user input handling, conditional branching using the Switch activity, and arithmetic operations. It also illustrated how basic programming logic can be applied within UiPath to build interactive automation workflows. This foundational understanding is essential for building more complex RPA solutions in the future.

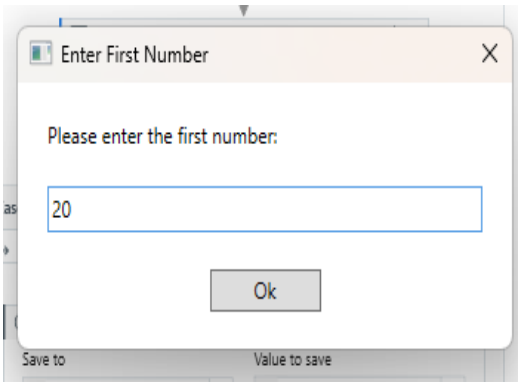
RPA PRACTICAL 2 OUTPUTS

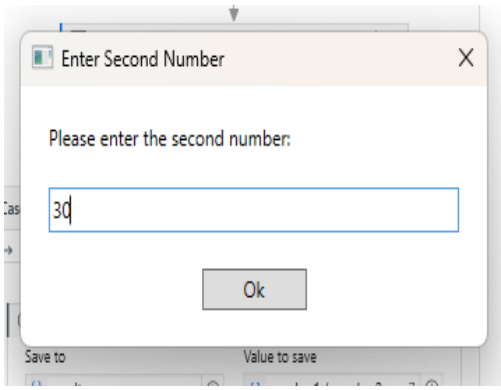


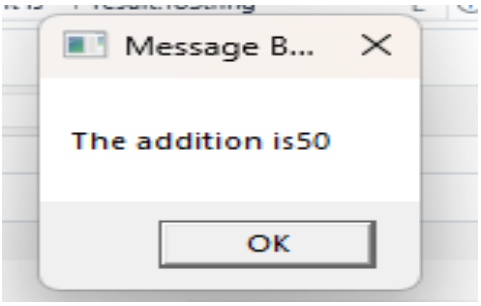

Switch
Expression *
operation
Case Add
Body
Assign
Save to
Value to save
result
number1 + number2
Message Box
Text *
"The addition is" + result.ToString
Case Subtract
Body
Assign
Save to
Value to save
result
number1 - number2
Message Box
Text *
"The subtraction is" + result.ToString


Case Multiply
Body
Assign
Save to
Value to save
result
number1 * number2
Message Box
Text *
"The result is " + result.ToString
Case Divide
Body
Assign
Save to
Value to save
result
number1 / number2
Message Box
Text *
"The result is" + result.ToString
Add case
Default
Drop activity here


Select Operation
Choose an operation: Add, Subtract, Multiply, or Divide.
Add
Ok
Save to
Value to save


Enter First Number
Please enter the first number:
20
Ok
Save to
Value to save


Enter Second Number
Please enter the second number:
30
Ok
Save to
Value to save


Message B...
The addition is50
OK

Experiment 3

Title: Use of control flow using if-else statement and Switch statement.

Aim: To understand and implement control flow in UiPath Studio using conditional constructs like **If-Else** and **Switch** statements for decision-making in automation workflows.

Objectives:

1. To understand the concept of control flow in programming.
2. To learn how to use **If-Else** decision-making statements in UiPath.
3. To explore the use of the **Switch** activity for multiple conditional branches.
4. To apply control flow statements in a sample UiPath automation project.
5. To observe and analyze the behavior of workflows based on different input conditions.

Theory:

What is Control Flow?

Control flow is the order in which individual statements, instructions, or function calls are executed in a program or workflow. In Robotic Process Automation (RPA), controlling the flow of execution is crucial for handling logic-based scenarios. UiPath Studio provides various control flow mechanisms that allow automation to behave differently based on conditions.

Two of the most commonly used control structures in UiPath are:

1. If-Else Statement

The **If** activity is used to represent a decision-making structure that evaluates a condition. If the condition is true, one set of actions is executed; otherwise, an alternative set of actions is executed.

Syntax in UiPath:

If (condition)

Then → Execute this sequence

Else

→ Execute this sequence

Example:
Check whether a number is even or odd. If the number modulo 2 is zero, it’s even; else, it’s odd.

2. Switch Statement

The **Switch** activity is used when there are multiple possible values that a variable might take, and different actions need to be executed depending on the value.

By default, Switch works with **Integer**, **String**, or **Enum** data types. It evaluates an expression and matches it with one of several predefined **cases**. If no case matches, an optional **Default** path is executed.

Syntax in UiPath:

Switch(expression)

Case "A" → Action A

Case "B" → Action B

Default → Default Action

Example:
Based on the day of the week entered by the user (e.g., "Monday", "Tuesday"), perform specific tasks.

Why Use Control Flow in UiPath?

Control flow allows the robot to:

- Make logical decisions.
- Perform different tasks based on conditions.
- Handle real-time user inputs, API responses, or data-driven scenarios.
- Enhance flexibility and intelligence in automation.

Both **If-Else** and **Switch** allow the automation process to mimic real-world decision-making and branching logic.

Program Input and Output :

Program 1: If-Else

Input:

User enters: -10

Output:

Message Box: Negative

Program 2: Switch

Input:

User enters: Spanish

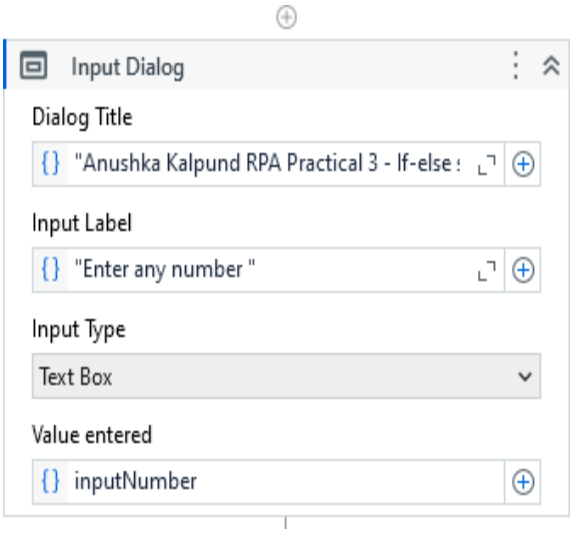
Output:

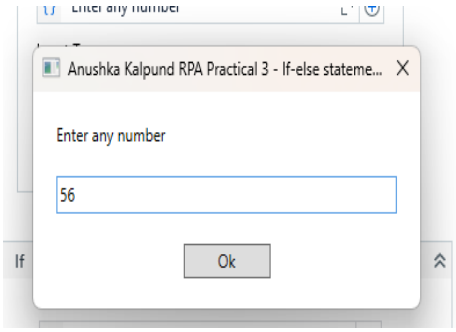
Message Box: ¡Hola!

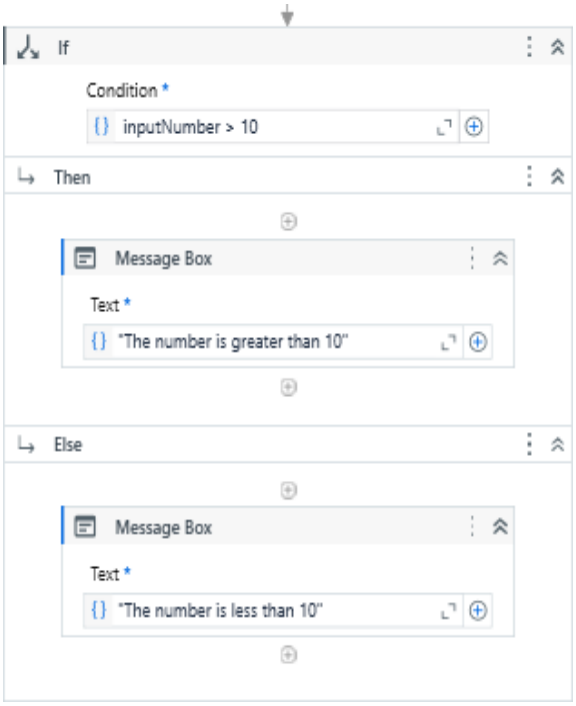
Conclusion :

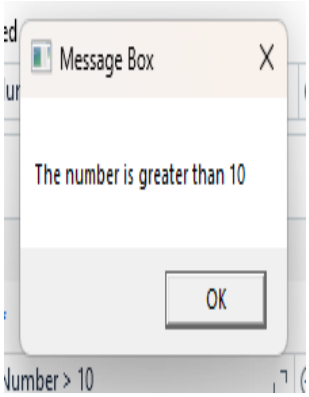
Using control flow mechanisms like **If-Else** and **Switch** statements in UiPath Studio allows automation developers to build intelligent, decision-based workflows. These constructs help the robot choose between different execution paths based on logic and user input, making the automation flexible, dynamic, and closer to real-world human decisions. Mastering control flow is essential for creating effective and robust automation solutions.

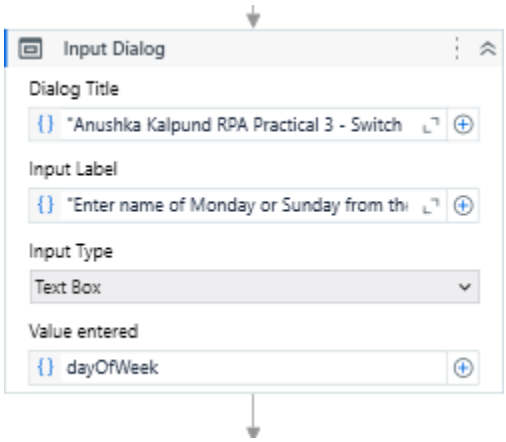
RPA PRACTICAL 3 OUTPUTS

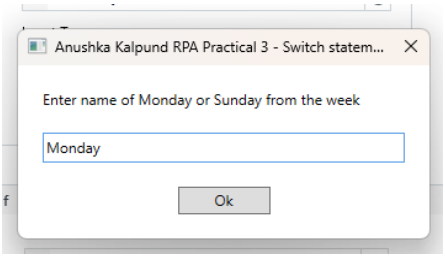


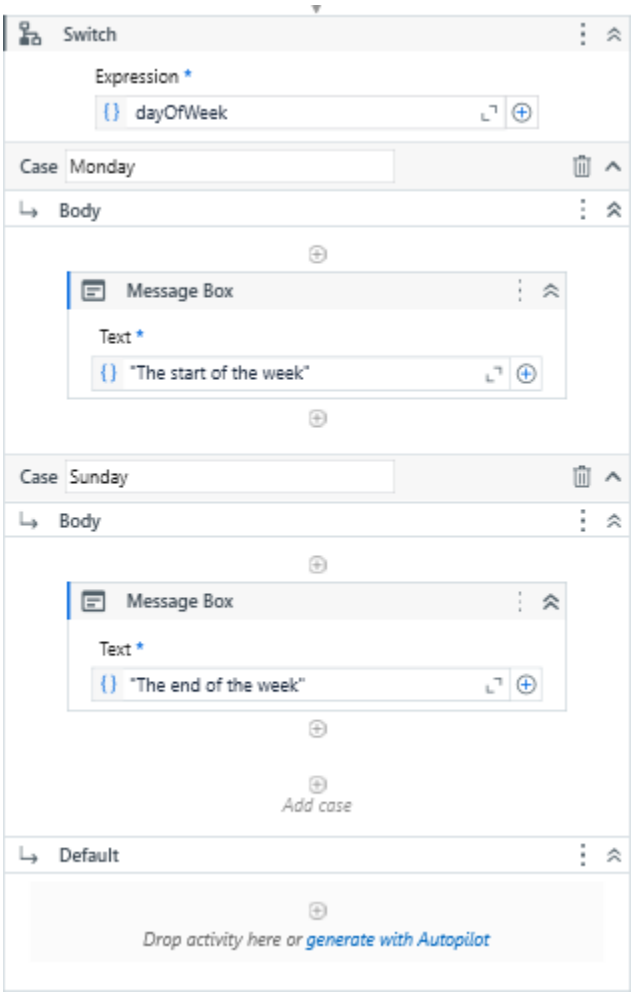


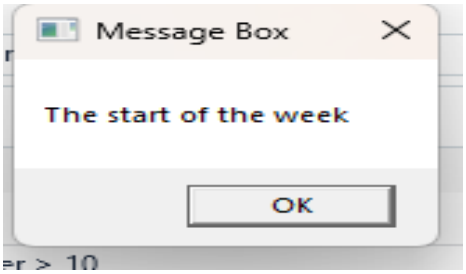





Input Dialog
Dialog Title
"Anushka Kalpund RPA Practical 3 - Switch
Input Label
"Enter name of Monday or Sunday from th
Input Type
Text Box
Value entered
dayOfWeek


Anushka Kalpund RPA Practical 3 - Switch statem...
Enter name of Monday or Sunday from the week
Monday
Ok


Switch
Expression *
dayOfWeek
Case Monday
Body
Message Box
Text *
"The start of the week"
Case Sunday
Body
Message Box
Text *
"The end of the week"
Add case
Default
Drop activity here or generate with Autopilot


Message Box
The start of the week
OK

Experiment 4

Title: Read text file and write the contents in another file. Extract contents of text file in message box.

Aim: To automate the process of reading a text file, displaying its contents in a Message Box, and writing the contents into another text file using UiPath Studio.

Objectives:

1. To understand how to use UiPath Studio for file handling operations.
2. To read the contents of a text file using the **Read Text File** activity.
3. To extract and display the contents using the **Message Box** activity.
4. To write the read content into another file using the **Write Text File** activity.
5. To build a reusable and simple automation workflow for basic file operations.

Theory:

Robotic Process Automation (RPA) enables software robots to mimic human interactions with digital systems to perform repetitive tasks. One of the most common tasks in automation is handling files — such as reading from or writing to text files — which is widely used in data processing, logging, reporting, and system integration.

UiPath Studio offers pre-built activities that allow users to manipulate files without writing traditional code. In this scenario, we are automating three actions:

1. Reading a Text File
   UiPath's Read Text File activity reads the full content of a specified .txt file and stores it as a string variable. This activity requires the full path to the input file and returns the entire content, regardless of its length.

2. Displaying the Content in a Message Box
   The Message Box activity is used to display the contents of the string variable in a popup dialog box. This is useful for validation, debugging, or simply showing the output to the user during runtime.

3. Writing the Text to Another File
   The Write Text File activity is used to save content to a new or existing file. It takes the content string and the output file path as input parameters.

Importance of File Handling in Automation

File operations are essential in RPA workflows because they facilitate:

- Integration with other systems (e.g., reading configuration files).
- Extraction and transformation of data for reports or logs.
- Communication between different stages of automation through intermediate files.

This project demonstrates how UiPath can handle such scenarios quickly and reliably with just a few drag-and-drop actions, making automation accessible even for non-developers.

UiPath Program Workflow

Activities Used:

- Read Text File
- Message Box
- Write Text File

Program Input and Output :

Input

A text file named input.txt with the following content:

"Hello this is Anushka and you are reading my file contents"

Output

1. A **Message Box** appears displaying:

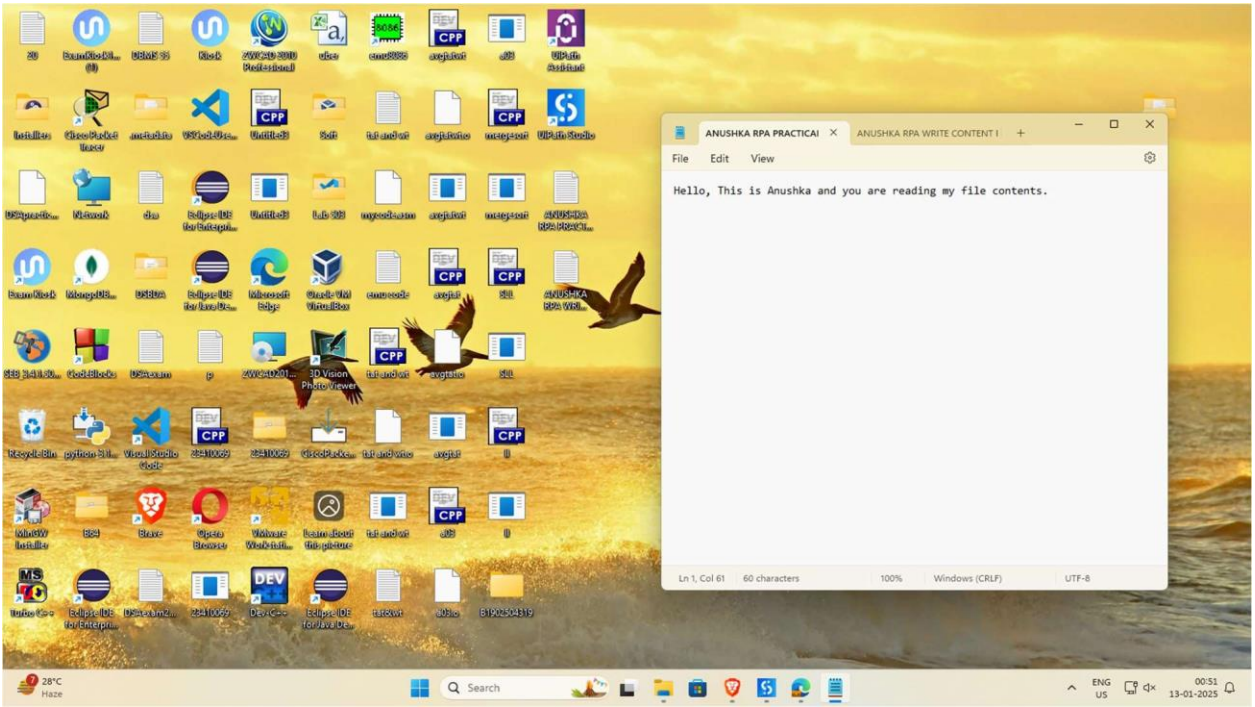
   "Hello this is Anushka and you are reading my file contents"

2. A new text file output.txt is created with the **same content** as input.txt.

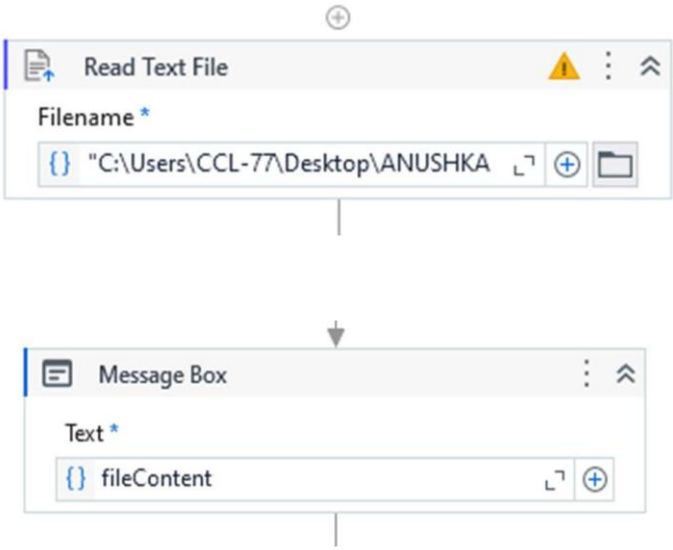
Conclusion :

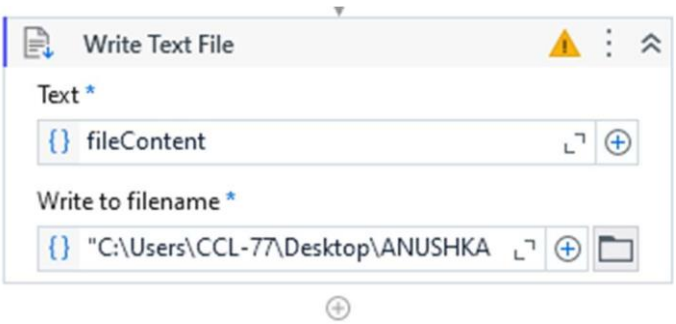
The automation was successfully implemented using UiPath Studio to read the contents of a text file, display them in a message box, and write them into a new file. This simple use case demonstrates the efficiency of UiPath for handling common file I/O operations. Such workflows are fundamental in automating document processing, log generation, and data migration tasks in real-world enterprise environments.

A file with some contents on it :

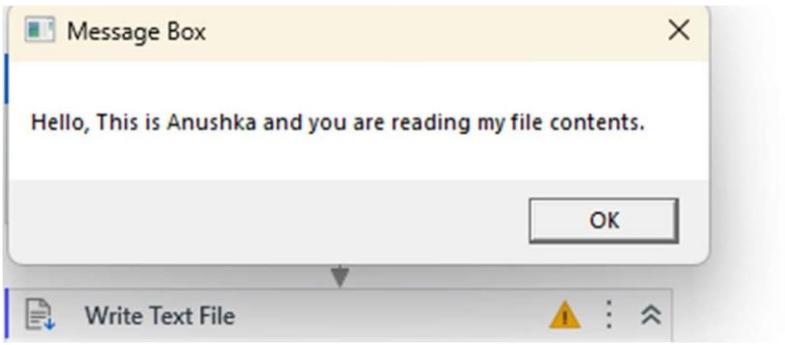


ACTIVITIES ON UiPath

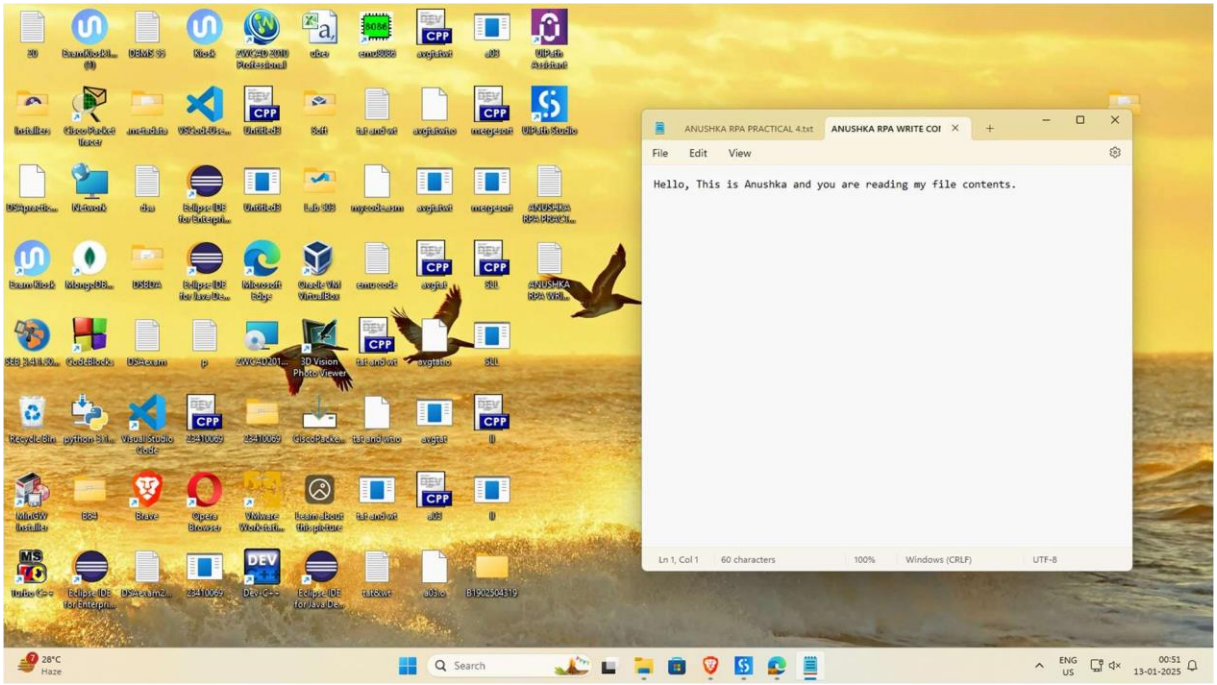




OUTPUT AFTER RUNNING :



Empty File is now written with the existing file content



Experiment 5

Title: Excel file automation/ text file automation.

Aim: To automate data processing tasks using Excel and text files in UiPath Studio by performing reading, writing, and manipulation operations through RPA workflows.

Objectives:

1. To learn how to use UiPath activities for handling Excel and text files.
2. To automate the reading of data from an Excel spreadsheet or text file.
3. To write or update specific content in Excel or text files using UiPath.
4. To improve task efficiency by eliminating manual file handling processes.

Theory:

Robotic Process Automation (RPA) has become an essential tool for automating repetitive and rule-based tasks. One of the most common use cases of RPA is **automating file-based operations**, such as interacting with **Excel spreadsheets** and **text files**. UiPath Studio provides built-in activities that make automating such file operations easy, reliable, and efficient.

Excel Automation in UiPath

Excel files are widely used in both small-scale and enterprise-level environments for storing structured data like reports, financial records, employee data, etc. Manually interacting with Excel can be time-consuming and error-prone, which makes Excel automation a valuable RPA use case.

UiPath offers two main methods to work with Excel:

1. **Workbook Activities**
    - These do not require Microsoft Excel to be installed on the machine.
    - Activities include:
        - Read Range (Workbook)
        - Write Range (Workbook)
        - Read Cell, Write Cell, etc.
2. **Excel Activities** (Require Excel to be installed)

- These use the Excel application and provide more flexibility.
- Activities include:
    - Excel Application Scope
    - Read Range
    - Write Range
    - Append Range, Delete Range, etc.

Excel automation allows users to read data from cells or ranges, write or append new data, format cells, create new sheets, and more.

Text File Automation in UiPath

Text files (.txt) are used to store unstructured or semi-structured data. UiPath provides simple and effective activities to read from and write to text files:

- Read Text File: Reads the entire content of a file and stores it in a string variable.
- Write Text File: Writes content to a file, replacing existing content.
- Append Text: Adds new content to the end of an existing text file without deleting the current content.

Text file automation is often used in logging, report generation, and data extraction processes.

Common Use Cases

- Reading Excel data to input into web forms.
- Generating reports by writing data to new Excel or text files.
- Data migration between systems using intermediate files.
- Summarizing logs and saving them into .txt or .xlsx format.

Program Input and Output :

Input (Excel Sheet):

Filename: SampleData.xlsx
Sheet1 Content:

Name	Age	Department
Alice	25	HR
Bob	30	IT

Output:

A new sheet named Output is created in the same file containing the same data.

Conclusion :

UiPath Studio simplifies the automation of file operations such as reading from and writing to Excel and text files. With just a few activities, developers can create robust automation workflows that streamline business processes. Excel and text file automation significantly reduces manual effort and enhances productivity in both individual and organizational environments.

Use Excel File

Excel file *
{} "InputFile.xlsx"
Reference as
Excel
✓ Save changes ✓ Create if not exists
Read formatting Same as project
Template file

Do

Read Range

Range *
{} Excel.Sheet("Sheet1")
✓ Has headers ✓ Visible rows only
Save to
{} dt_ExcelData

Write DataTable to Excel

What to write *
{} Excel.Sheet("Sheet1").Range("A1:F5")
Destination *
{} Excel.Sheet("Sheet2")
Append ✓ Exclude headers

Insert Chart

Chart category * Chart type *
Bar Clustered Bar
Data range *
{} Excel.Sheet("Sheet1").Range("A2:F5")
Insert into sheet *
{} Excel.Sheet("Sheet1")
Save chart to
{} InsertedChart

Insert Chart

Chart category * Chart type *
Pie Pie
Data range *
{} Excel.Sheet("Sheet2").Range("A2:F5")
Insert into sheet *
{} Excel.Sheet("Sheet2")
Save chart to
{} InsertedChart

Message Box

Text *
{} "Data successfully represented into Bar and

Experiment 6

Title: PDF Automation.

Aim: To automate the process of extracting and manipulating data from PDF files using UiPath Studio, thereby reducing manual effort and increasing efficiency in document processing tasks.

Objectives:

1. To understand and implement PDF automation using UiPath Studio.
2. To learn how to use PDF-related activities like `Read PDF Text`, `Read PDF with OCR`, and `Extract PDF Page Range`.
3. To extract specific text or data from structured or unstructured PDF files.
4. To automate data entry or storage based on the extracted content.
5. To improve the speed, accuracy, and consistency of document processing tasks.

Theory:

Introduction to PDF Automation

PDF (Portable Document Format) files are commonly used in businesses for reports, invoices, resumes, forms, and contracts. Extracting data manually from a large number of PDF documents is tedious, error-prone, and time-consuming. **UiPath Studio** offers a suite of activities under its **PDF package** to automate the extraction and handling of PDF data, allowing users to build efficient workflows for document management.

PDF Automation is a subset of **Document Understanding** and is especially useful in use-cases like invoice processing, data validation, customer form analysis, and archiving information from structured/unstructured PDF files.

Types of PDFs

There are two major types of PDFs:

1. **Text-Based PDFs** – These contain selectable and searchable text.

2. **Image-Based PDFs (Scanned PDFs)** – These are images of text and require OCR (Optical Character Recognition) to extract content.

UiPath provides support for both types using different activities and OCR engines.

Key PDF Activities in UiPath Studio

UiPath offers PDF automation activities via the **UiPath.PDF.Activities** package (which can be installed from the Manage Packages window).

Common PDF activities include:

- **Read PDF Text**: Reads the text content from a text-based PDF file.
- **Read PDF with OCR**: Reads scanned PDFs using OCR engines like Tesseract, Google Cloud Vision, or Microsoft OCR.
- **Extract PDF Page Range**: Extracts specified pages from a PDF into a new file.
- **Export PDF Page As Image**: Converts a PDF page into an image format.

These activities can be combined with data manipulation actions, file system operations, or even Excel/database operations for advanced automation workflows.

OCR in PDF Automation

For scanned or image-based PDFs, UiPath uses **OCR (Optical Character Recognition)** to convert the image into editable and searchable text. OCR engines like **Tesseract**, **Microsoft OCR**, or **Google OCR** must be configured to read and extract accurate data.

OCR accuracy depends on:

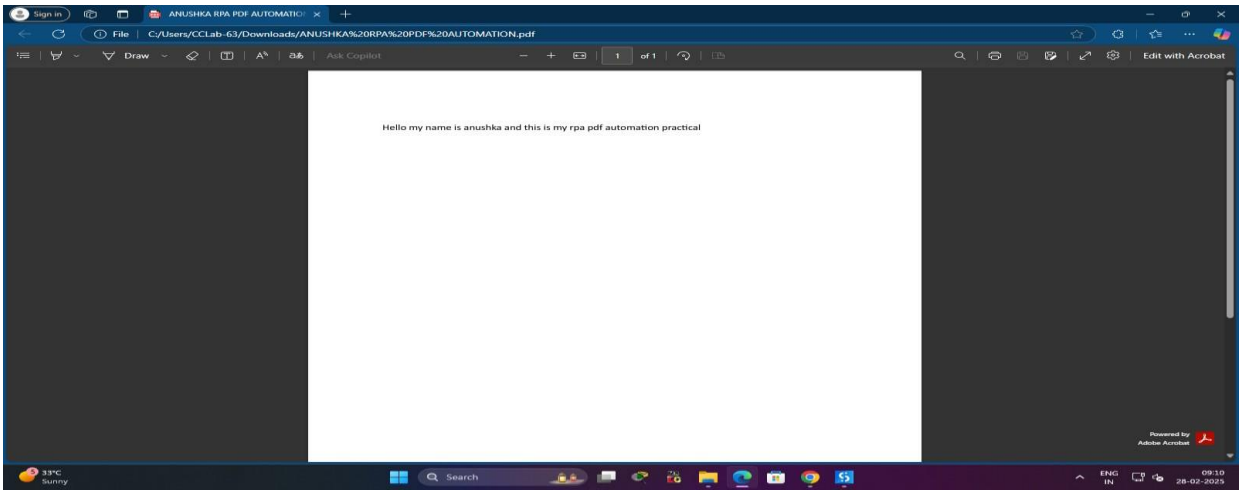
- Image resolution and quality
- Language used in the document
- Structure and formatting of the document

Conclusion :

PDF Automation using UiPath Studio is a powerful way to handle large volumes of document processing tasks efficiently. By utilizing the built-in PDF activities and OCR technology, users can automate the extraction of text and data from various types of PDFs — saving time, improving accuracy, and enhancing workflow productivity. This automation has applications across multiple domains including finance, HR, healthcare, and logistics, making it an essential skill for modern RPA developers.

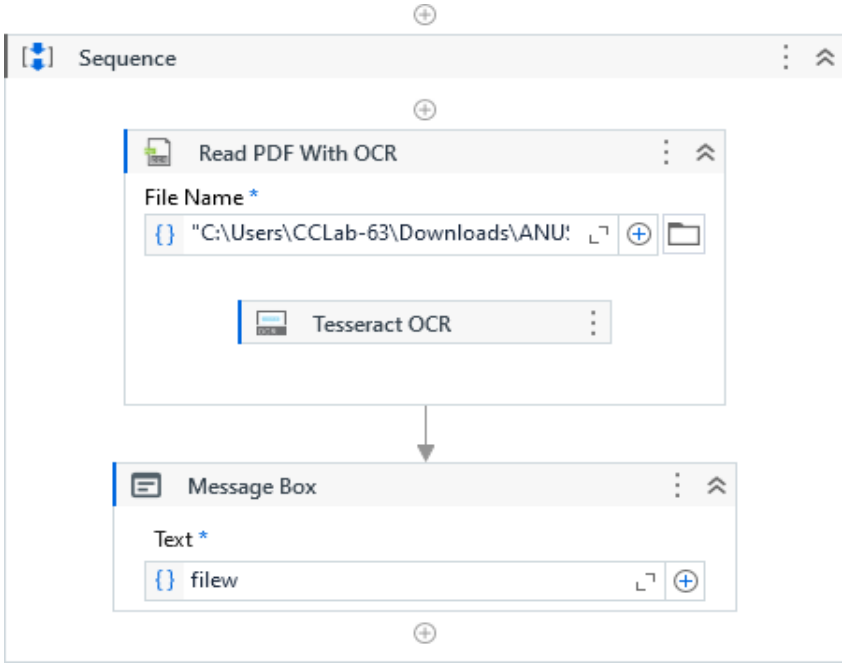
PDF AUTOMATION

1. Create a PDF file on your PC and add some content in it.



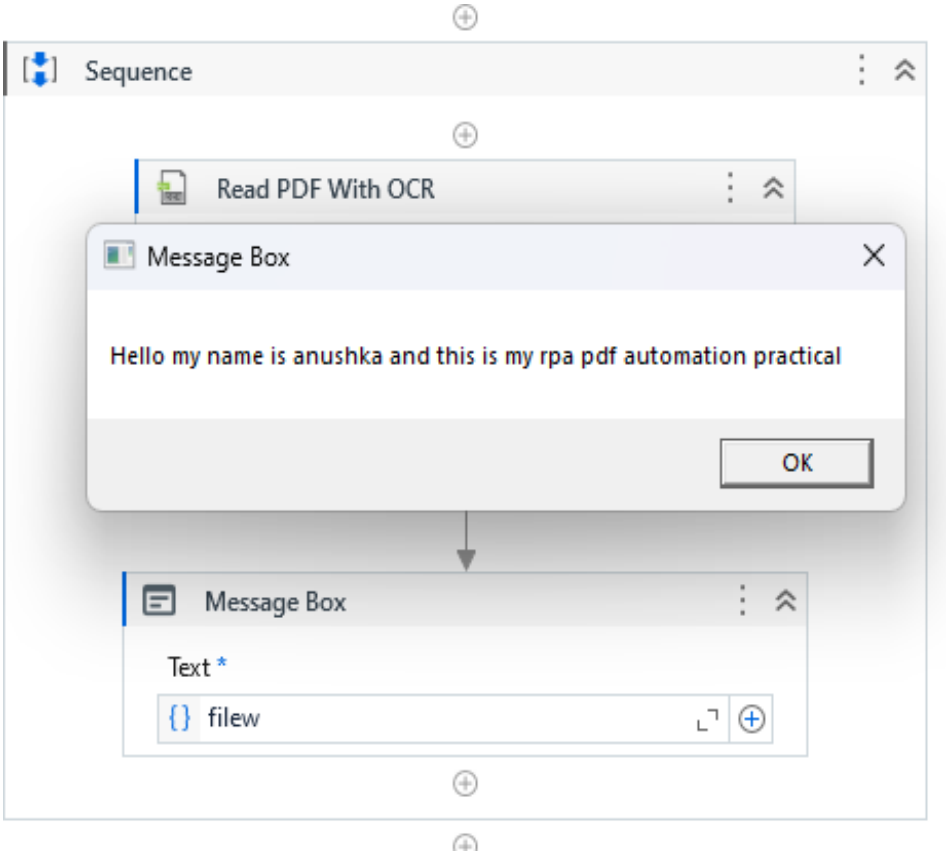
2. Open UiPath Studio and drag the following activity :

"Read PDF with OCR"



3. Add the PDF files path, a varaible and a message box.

4. Run the activity.


Sequence
Read PDF With OCR
Message Box
Hello my name is anushka and this is my rpa pdf automation practical
OK
Message Box
Text *
filew

Experiment 7

Title: Web Scraping.

Aim: To extract structured data from a website using the **Web Scraping** capabilities of **UiPath Studio** and automate the process of collecting and storing online data.

Objectives:

1. To understand the concept of web scraping and its real-world applications.
2. To explore UiPath Studio's **Data Scraping Wizard** for automating web data extraction.
3. To build a workflow that navigates to a website and extracts relevant data such as names, prices, links, or descriptions.
4. To store the scraped data in a structured format such as a CSV or Excel file.
5. To test and validate the automation workflow with real-time website data.

Theory:

What is Web Scraping?

Web Scraping is the process of automatically extracting data from websites. It is commonly used in industries for market analysis, competitive pricing, product research, job listings, and more. Instead of manually copying information from web pages, web scraping tools simulate human interaction with a browser to navigate, identify, and retrieve specific content.

In traditional programming, web scraping is done using libraries like BeautifulSoup (Python) or Jsoup (Java). However, with UiPath Studio, web scraping becomes more accessible, as it provides a no-code/low-code environment and built-in tools to automate the entire process without deep programming knowledge.

Web Scraping in UiPath Studio

UiPath Studio offers a specialized Data Scraping Wizard that helps users extract structured data from web pages, such as:

- Tables
- Lists of items (e.g., product names, prices)
- Links
- Images
- Text fields

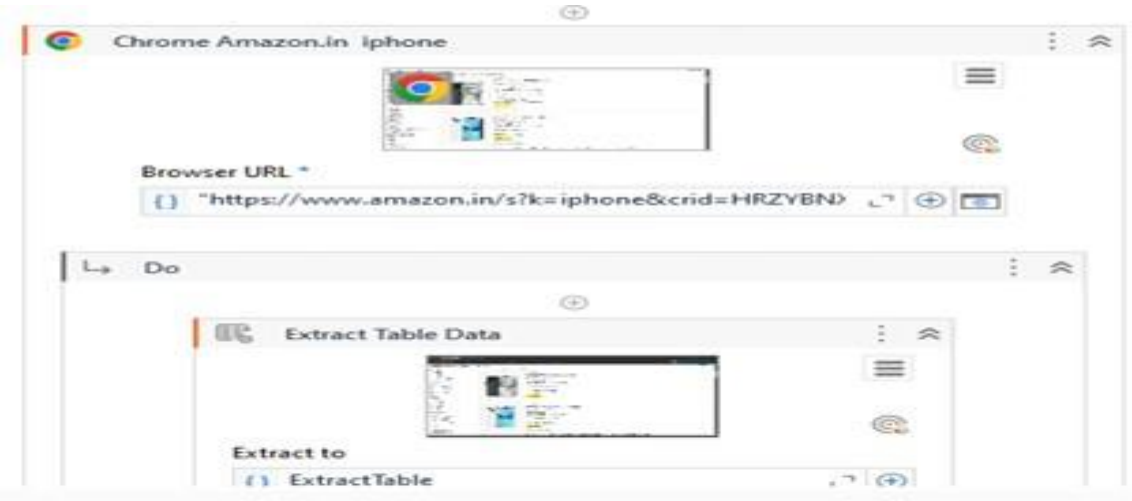
UiPath leverages selectors and UI automation under the hood to understand the structure of the HTML page and extract repeating elements, often seen in e-commerce sites, directories, job portals, etc.

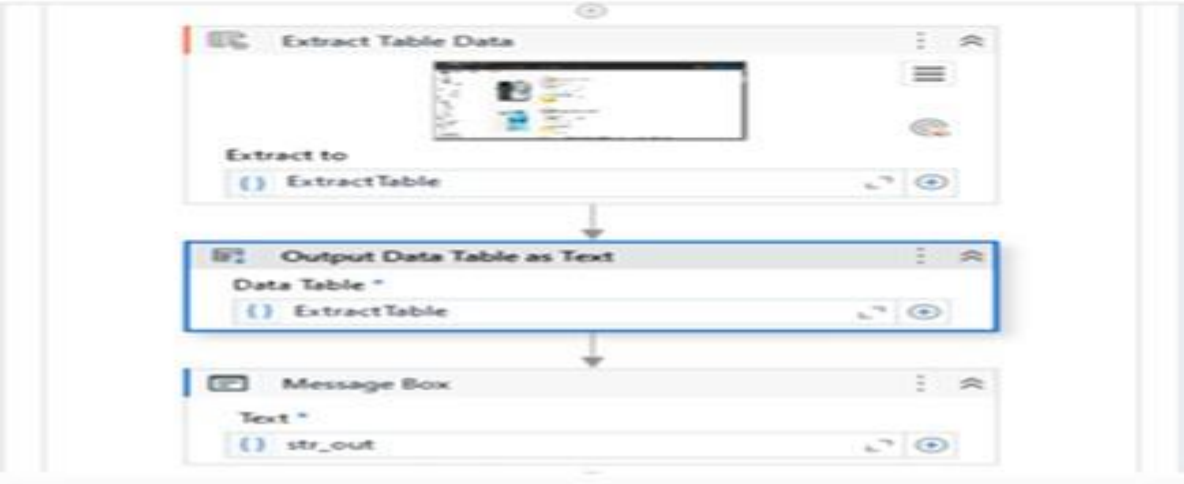
Key Features:

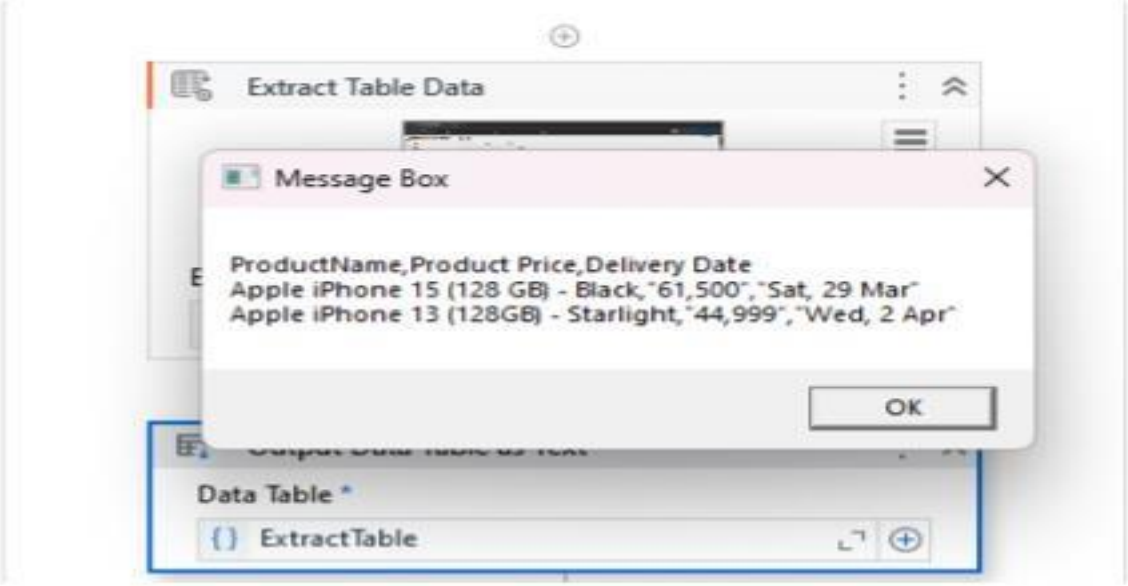
- Easy-to-use Wizard for extracting structured data.
- Ability to scrape multiple pages (pagination).
- Exporting data to Excel, CSV, or DataTable variables.
- Built-in browser support (Chrome, Edge, Firefox).

Conclusion :

Web scraping using UiPath Studio provides a simple and powerful approach to automate the collection of structured data from websites. With minimal coding, users can build workflows to extract valuable information from online sources and use it for business intelligence, reporting, or analysis. UiPath's Data Scraping Wizard significantly reduces the technical complexity of scraping and empowers both technical and non-technical users to harness web data efficiently.


Chrome Amazon.in iphone
Browser URL *
"https://www.amazon.in/s?k=iphone&crid=HRZYBN
Do
Extract Table Data
Extract to
ExtractTable


Extract Table Data
Extract to
ExtractTable
Output Data Table as Text
Data Table *
ExtractTable
Message Box
Text *
str_out


Extract Table Data
Message Box
ProductName,Product Price,Delivery Date
Apple iPhone 15 (128 GB) - Black,"61,500","Sat, 29 Mar"
Apple iPhone 13 (128GB) - Starlight,"44,999","Wed, 2 Apr"
OK
Data Table *
ExtractTable

Experiment 8

Title: Email Automation.

Aim: To design and implement an automation workflow in UiPath Studio that reads, sends, or processes emails automatically using configured email services like Outlook or SMTP/IMAP.

Objectives:

1. To understand how UiPath Studio interacts with email applications and servers.
2. To configure email services such as Outlook, Gmail (via SMTP/IMAP), or Exchange in UiPath.
3. To automate the process of reading unread emails, extracting data, and sending emails with or without attachments.
4. To develop and execute a sample email automation project using UiPath activities.
5. To demonstrate the efficiency and usefulness of automating routine email tasks.

Theory:

Email Automation is one of the most common and powerful use cases of Robotic Process Automation (RPA). It refers to the automatic handling of email-related tasks such as sending emails, reading inbox content, downloading attachments, responding to specific messages, or even sorting and archiving emails based on certain criteria.

☐ **UiPath Studio and Email Automation**

UiPath Studio provides dedicated **email activities** under packages such as:

- **UiPath.Mail.Activities**
- **UiPath.Outlook.Activities**
- **UiPath.GSuite.Activities**

These packages allow seamless integration with email platforms like:

- Microsoft Outlook (Desktop Application)
- Gmail via SMTP, IMAP, or GSuite
- Microsoft Exchange Server
- Office 365

These activities include:

- **Send Outlook Mail Message**
- **Get Outlook Mail Messages**
- **Send SMTP Mail Message**

- **Send IMAP Mail Message**
- **Save Mail Attachments**
- **Filter Emails** using keywords, subjects, or sender info

☐ **Use Cases of Email Automation**

- **Customer Service**: Automating ticket responses or acknowledgments.
- **Invoice Management**: Extracting and downloading invoices from email attachments.
- **Reporting**: Automatically sending scheduled reports.
- **Notifications**: Sending alerts for task completions or system failures.

☐ **Email Protocols**

- **SMTP (Simple Mail Transfer Protocol)**: Used for sending emails.
- **IMAP (Internet Message Access Protocol)**: Used for reading emails from the server.
- **POP3**: Another protocol for reading emails, less commonly used in RPA.

☐ **Steps in Email Automation using UiPath**

1. **Install the required email activity packages** from the Manage Packages section.
2. **Configure the activity properties**, including server name, port number, email, password, and security (SSL/TLS).
3. **Use read/send activities** based on your requirement (e.g., reading unread emails from inbox, downloading attachments).
4. **Use loops and conditions** to process each mail as needed.
5. **Send automated responses** or forward emails using Send Mail activities.

☐ **Security Note**

For Gmail or other secure email providers, **app-specific passwords or OAuth2.0 authentication** might be required instead of plain email credentials due to enhanced security protocols.

Conclusion :

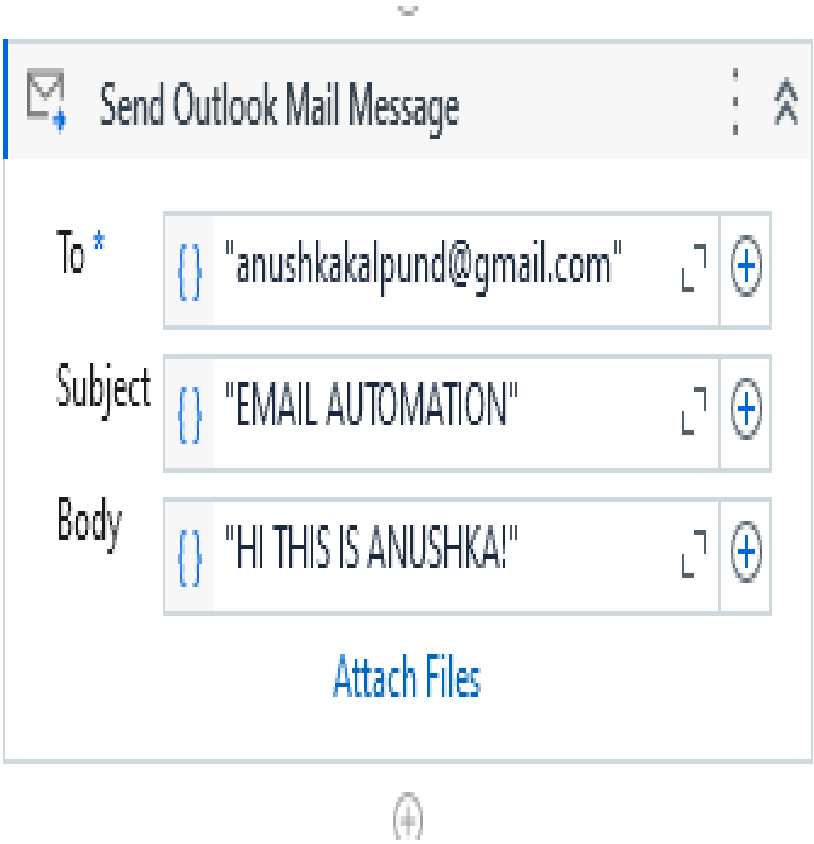
Email Automation using UiPath Studio significantly reduces the manual effort required for routine communication tasks. It is especially useful in enterprise environments for tasks like auto-responses, data extraction, and alert systems. By leveraging protocols like SMTP and IMAP, UiPath enables seamless integration with popular email providers. The successful execution of the workflow demonstrates the power and flexibility of RPA in handling email processes efficiently and securely.

EMAIL AUTOMATION

1. Login to your Outlook mail account.

2. Open UiPath Studio and add the following activity :

"Send Outlook Mail Message"



3. Write :

a) To : Other's email

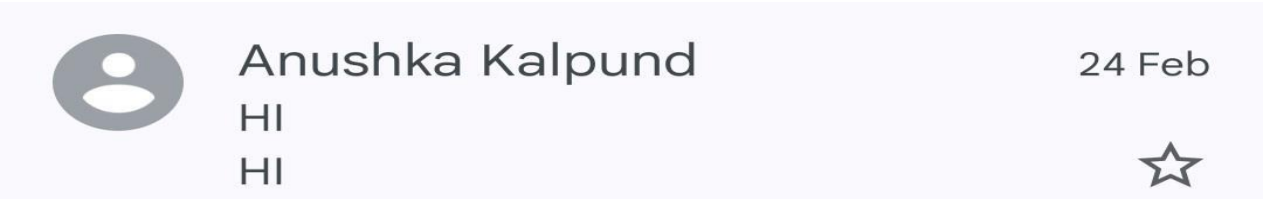
b) Subject : Email's Subjects

c) Body : Email's Body

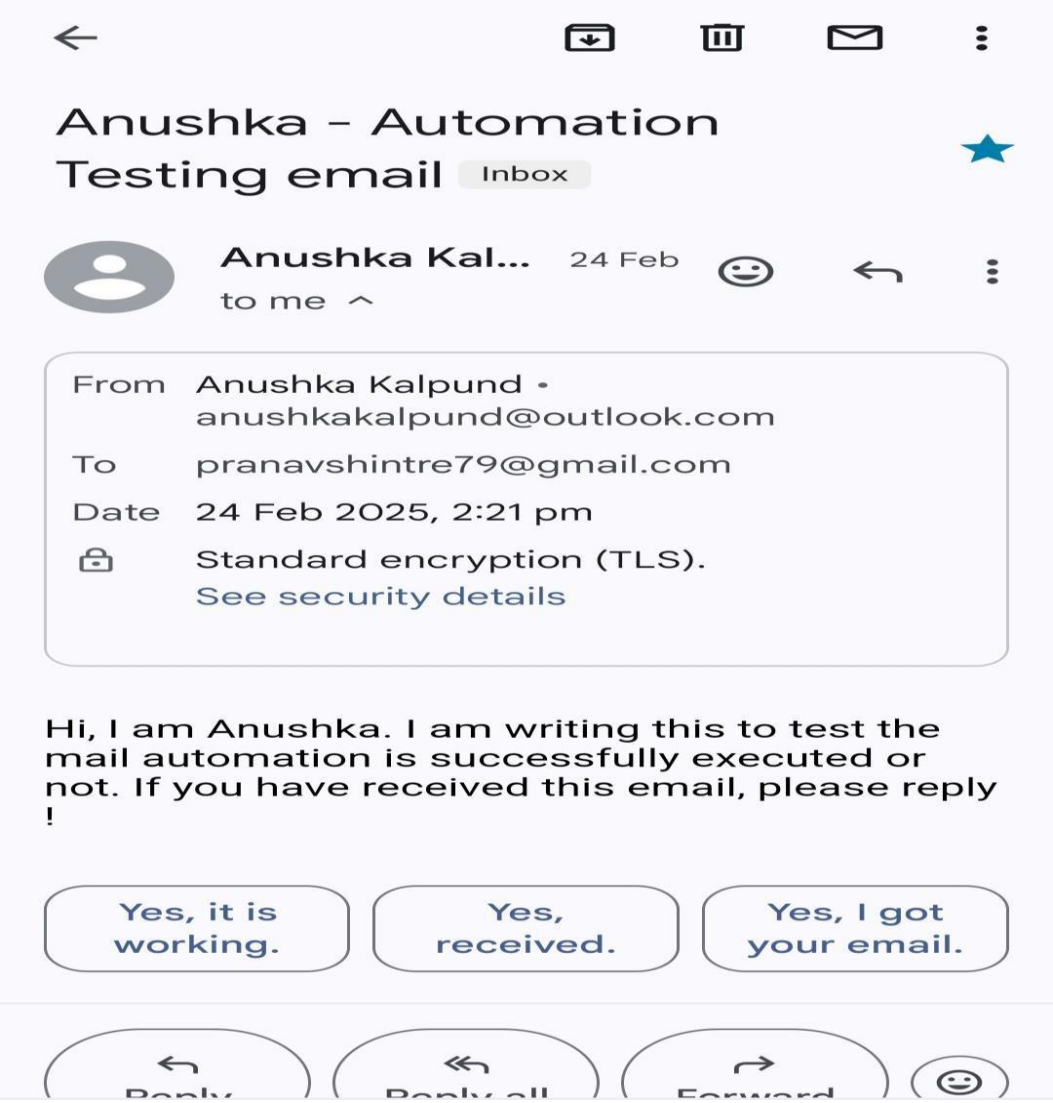
4. Run the Activity

5. Check your gmail account :

Email testing 1 :



Email testing 2 :



Experiment 9

Title: Keyboard and mouse automation.

Aim: To automate keyboard and mouse actions using UiPath Studio, such as clicking, typing, and navigating through user interfaces, to simulate human-computer interaction in repetitive tasks.

Objectives:

1. To understand and implement mouse automation using UiPath activities like `Click`, `Double Click`, and `Hover`.
2. To simulate keyboard input using activities like `Type Into`, `Send Hotkey`, and `Set Text`.
3. To create a sample workflow that combines both keyboard and mouse operations for UI automation.
4. To explore selector-based and image-based automation techniques for precise control.
5. To verify the automation on standard applications (e.g., Notepad, Calculator, or browser).

Theory:

Introduction to Keyboard and Mouse Automation

Keyboard and mouse automation refers to the process of programmatically replicating human interactions with a computer interface using software tools. UiPath Studio provides a powerful set of built-in activities to automate these types of interactions, allowing users to eliminate manual and repetitive tasks that involve UI-based navigation.

RPA tools like UiPath use two primary modes to interact with UI elements:

- **Selector-based automation**, which interacts with UI elements via their underlying XML-like structure.
- **Image or coordinate-based automation**, which relies on screen elements' visual appearance or positions.

Mouse Automation in UiPath

Mouse automation includes actions that simulate a human using a mouse, such as:

- **Click Activity**: Performs a single click (left or right) on a UI element.
- **Double Click Activity**: Simulates a double-click operation.
- **Right Click Activity**: Executes a context menu click.
- **Hover Activity**: Moves the cursor over a specific element without clicking.

- **Click Image / Click Text**: Used when traditional selectors fail (e.g., in Citrix or remote environments).

These activities require **selectors** to identify the target UI component. Selectors are XML fragments that uniquely identify UI elements based on properties like window title, class, text, etc.

Keyboard Automation in UiPath

UiPath provides several keyboard-related activities:

- **Type Into**: Simulates typing text into input fields.
- **Send Hotkey**: Sends function keys or key combinations like Ctrl+C, Alt+Tab, etc.
- **Set Text**: Directly sets text in a UI element (more reliable but may not simulate keystrokes).

These are essential for:

- Filling out forms.
- Sending keyboard shortcuts.
- Navigating menus.
- Opening or saving files using keyboard commands.

Real-Time Applications

Keyboard and mouse automation are used in scenarios like:

- Filling online registration forms.
- Logging into web portals.
- Copy-pasting data between Excel and ERP software.
- Opening Notepad, typing content, and saving it.
- Testing UI-based applications for QA purposes.

Conclusion :

UiPath Studio provides a robust platform for automating keyboard and mouse actions, enabling efficient human-like interaction with computer systems. This helps automate repetitive, rule-based tasks across various applications, enhancing productivity and reliability. By mastering these automation techniques, users can automate real-world tasks such as form-filling, data entry, and system navigation without manual intervention.

Input Dialog

Dialog Title

{} "Enter choice of automation"

Input Label

{} "Enter 1 for mouse and 2 for keyboard"

Input Type

Text Box

Value entered

{} UserChoice

Switch

Expression *

{} UserChoice

Case 1

Switch

Expression *

{} UserChoice

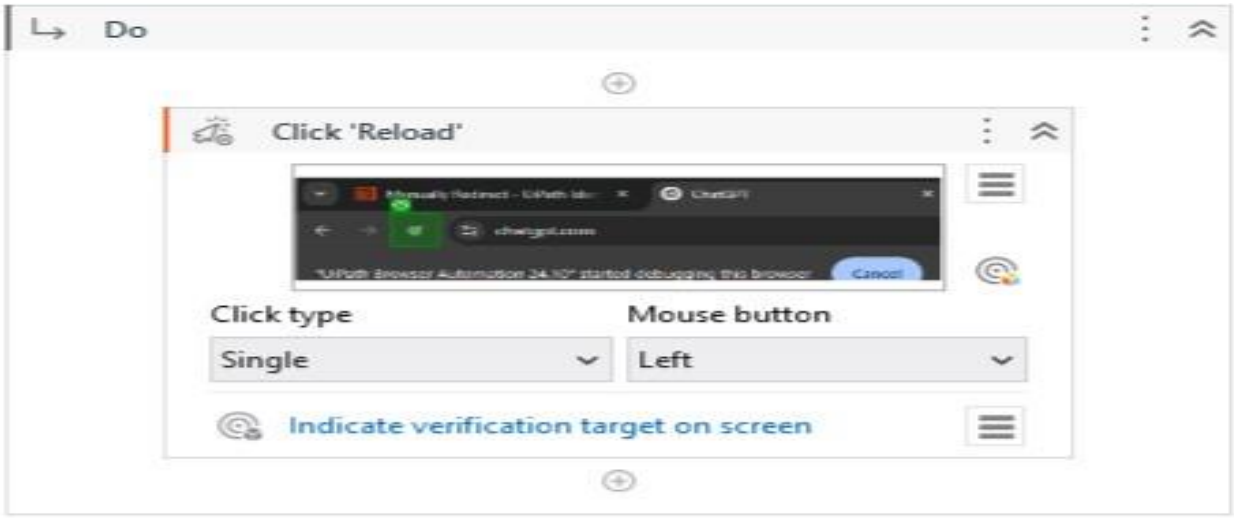
Case 1

Body

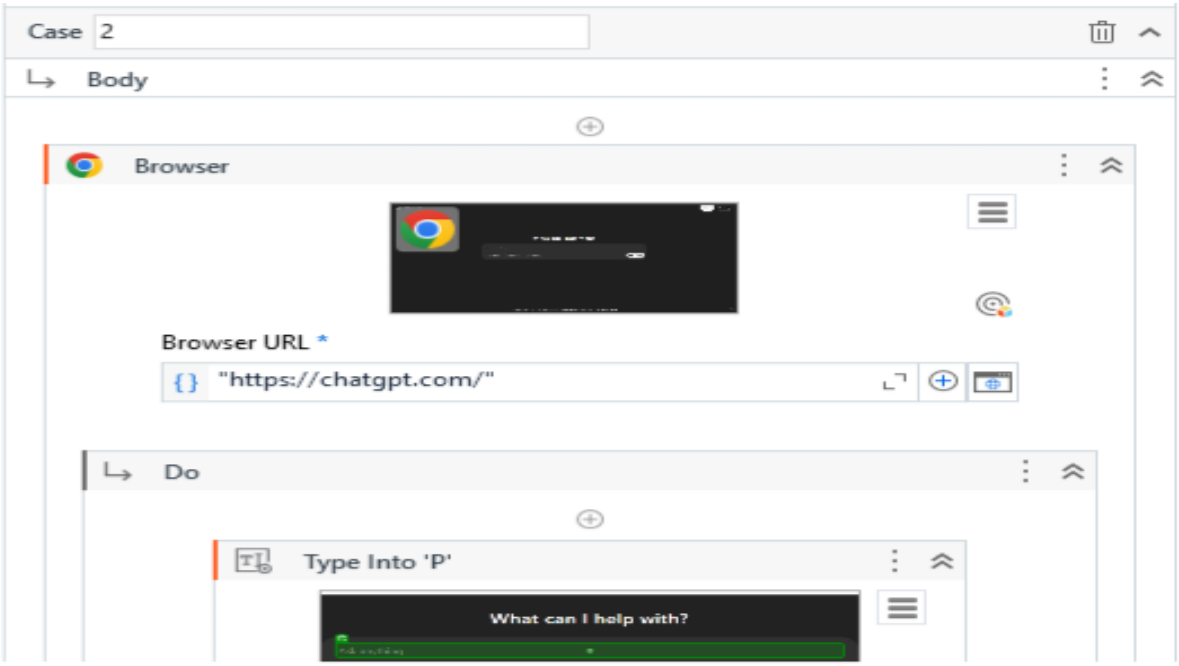
Browser

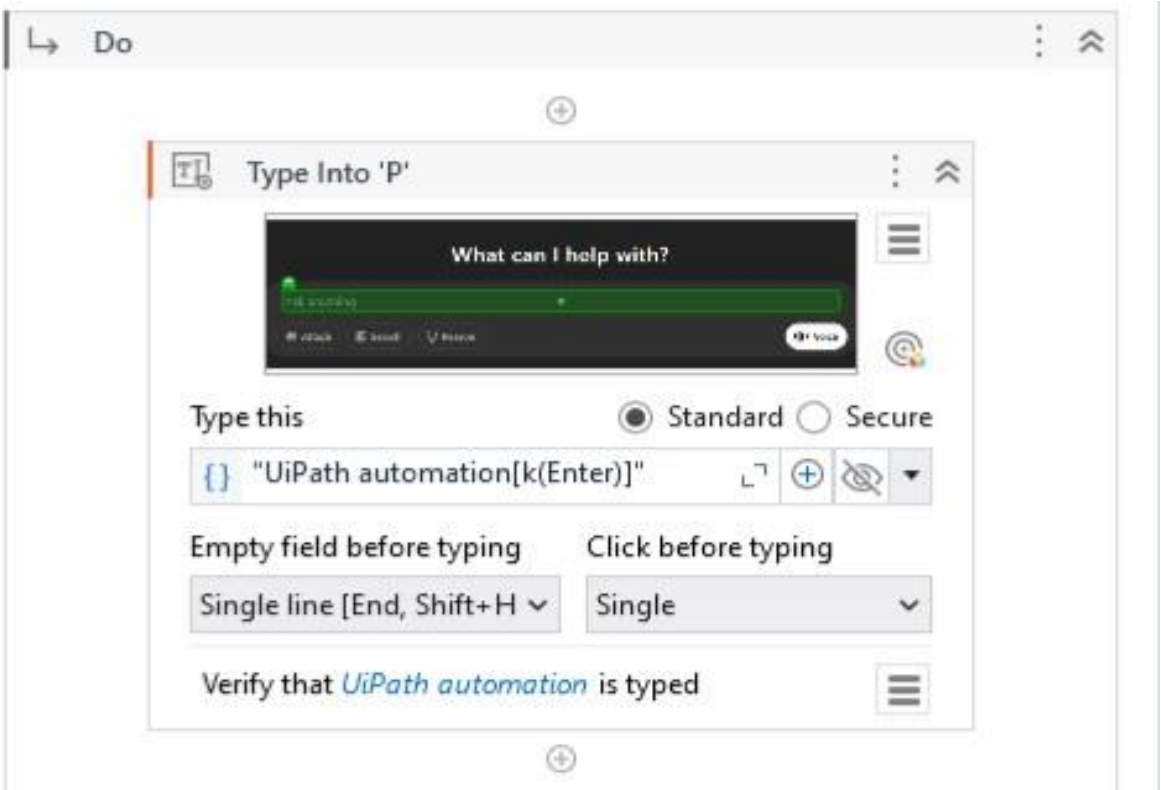
Browser URL *

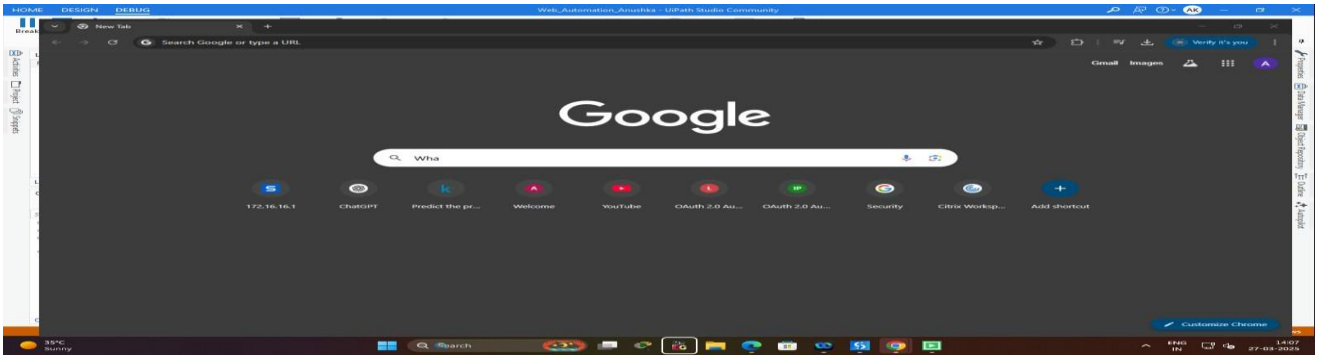
"https://chatgpt.com/"

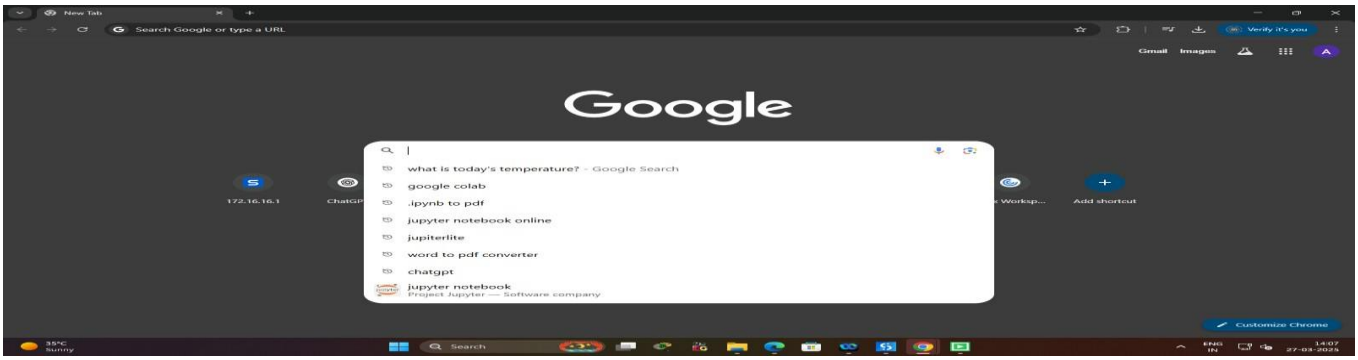

Do
Click 'Reload'
Click type
Mouse button
Single
Left
Indicate verification target on screen

se 2
Body


Case 2
Body
Browser
Browser URL *
"https://chatgpt.com/"
Do
Type Into 'P'
What can I help with?


Do
Type Into 'P'
What can I help with?
Type this
Standard
Secure
"UiPath automation[k(Enter)]"
Empty field before typing
Click before typing
Single line [End, Shift+H
Single
Verify that UiPath automation is typed


Google


Google
what is today's temperature? - Google Search
google colab
.ipynb to pdf
jupyter notebook online
jupiterlite
word to pdf converter
chatgpt
jupyter notebook
Project Jupyter — Software company

Experiment 10

Title: Sharable workflow.

Aim: To create a reusable and sharable workflow in UiPath Studio that can be integrated across multiple automation processes, enhancing modularity, collaboration, and efficiency in robotic process automation development.

Objectives:

1. To understand the concept and structure of reusable workflows in UiPath Studio.
2. To create a sharable workflow using a separate `.xaml` file.
3. To integrate the sharable workflow into a main process using the **Invoke Workflow File** activity.
4. To demonstrate modular programming principles in RPA.
5. To promote code reuse, reduce redundancy, and streamline development efforts.

Theory:

Introduction to Sharable Workflows

In Robotic Process Automation (RPA), especially when building complex automation solutions, code reusability is crucial. UiPath Studio allows developers to create sharable workflows — separate .xaml files — which encapsulate specific functionality or logic that can be reused in multiple other workflows. This modular approach not only improves the maintainability and scalability of RPA projects but also encourages best practices like separation of concerns and reduced duplication.

What is a Sharable Workflow?

A sharable workflow in UiPath refers to a reusable .xaml file that contains a well-defined set of actions or logic. It functions like a sub-program or function that can be invoked from one or more parent workflows using the Invoke Workflow File activity. Input and output arguments can be passed between workflows to ensure data exchange and flow control.

This modular approach is comparable to using functions or procedures in traditional programming languages like Java or Python. It enhances team collaboration, allows version control, and makes testing individual units easier.

Types of Reusability in UiPath:

1. Library Reusability: Create custom activities using Libraries and publish them as packages.
2. Workflow Reusability: Use .xaml files with input/output arguments to reuse within projects.
3. Snippet Reusability: Save frequently used blocks of code as snippets for drag-and-drop use.

Among these, workflow reusability using Invoke Workflow File is the most accessible and beginner-friendly method.

Advantages of Using Sharable Workflows:

- Modularity: Breaks down large automation into smaller, manageable components.
- Maintainability: Makes it easier to update and test a specific piece of logic without affecting the entire process.
- Team Collaboration: Multiple developers can work on different workflows simultaneously.
- Reduced Redundancy: Avoids rewriting the same logic across multiple projects.
- Enhanced Performance: Promotes organized and scalable automation design.

Real-Life Use Cases:

- Login modules shared across different automation bots.
- Email sending modules reused in multiple workflows.
- Data validation modules applied to different forms or sources.
- Excel reading/writing logic used by various bots.

Conclusion :

Sharable workflows in UiPath Studio empower developers to design modular, maintainable, and reusable automation components. By isolating specific logic into independent .xaml files and using the **Invoke Workflow File** activity, teams can build scalable RPA solutions that are easy to debug, upgrade, and share across projects. This approach not only reduces development time but also enforces best practices and promotes teamwork in enterprise automation projects.

STEP 1

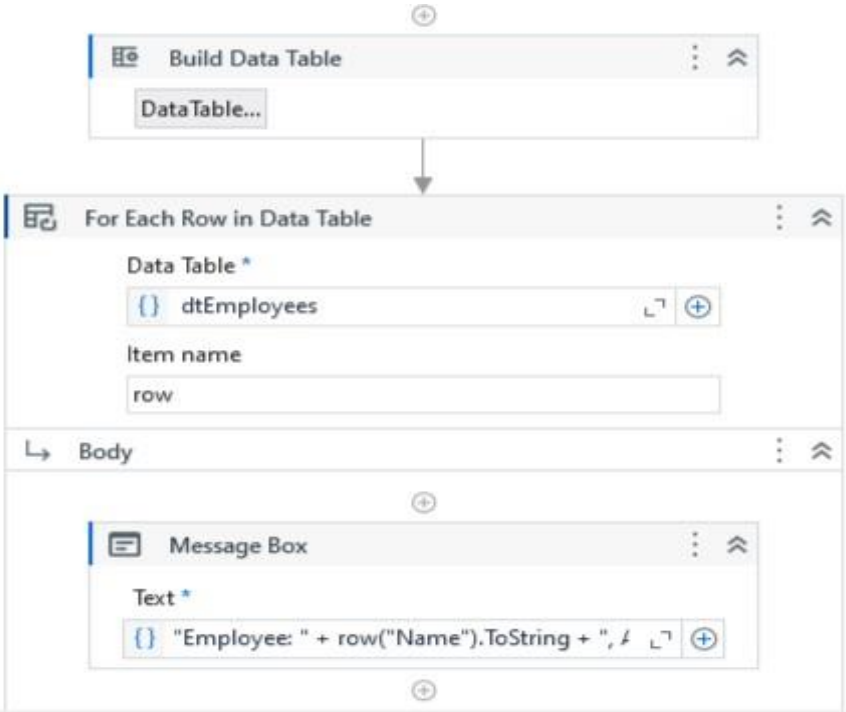
1. In the Activities Panel, search for “Build Data Table” activity.

2. Drag and drop Build Data Table into the Main.xaml workflow.

3. Click on "DataTable" and define columns (e.g., ID, Name, Age).

4. Add some sample data by clicking "Add Row" (e.g., 1, Anushka, 23).

5. Click OK and assign it to a variable (e.g., dtEmployees).



STEP 2

1. Drag For Each Row in Data Table activity.

2. Select the dtEmployees variable as input.

3. Inside the loop, use "Log Message" to print row values:

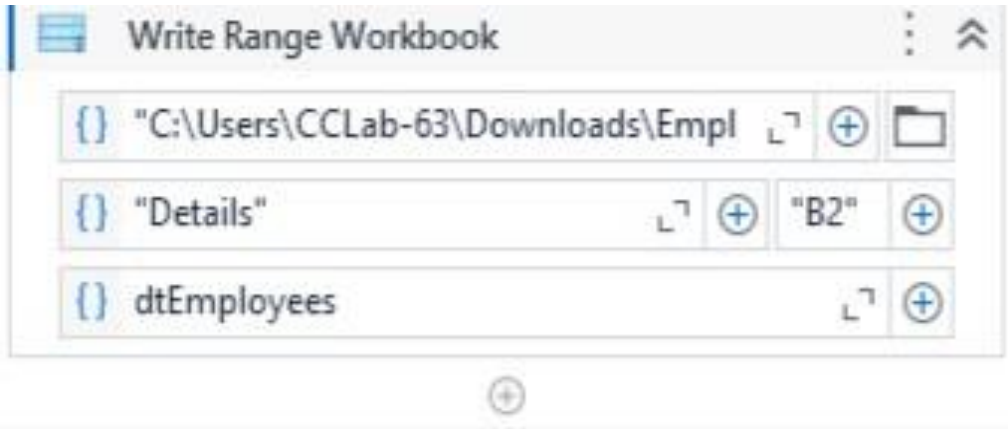
row("Name").ToString + " is " + row("Age").ToString + " years old"

STEP 3

1. Drag and drop "Write Range" activity from Excel package.

2. Connect it to an Excel file path (e.g., C:\Users\YourName\Documents\Data.xlsx).

3. Set the sheet name (e.g., "Employees").



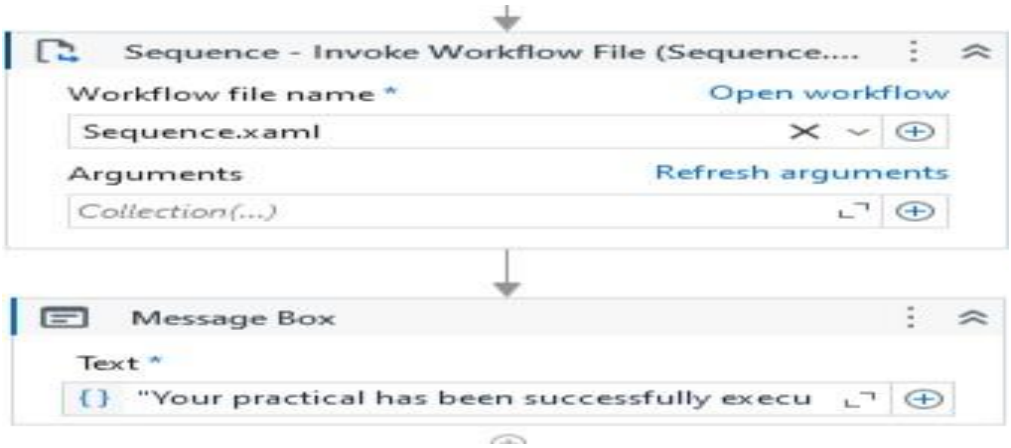
STEP 4

1. Create a new Sequence workflow (Right-click the project > Add > Sequence).

2. Move the data table logic into this sequence.

3. Use "Invoke Workflow File" to call this sequence in Main.xaml.

4. This makes the workflow modular and sharable.

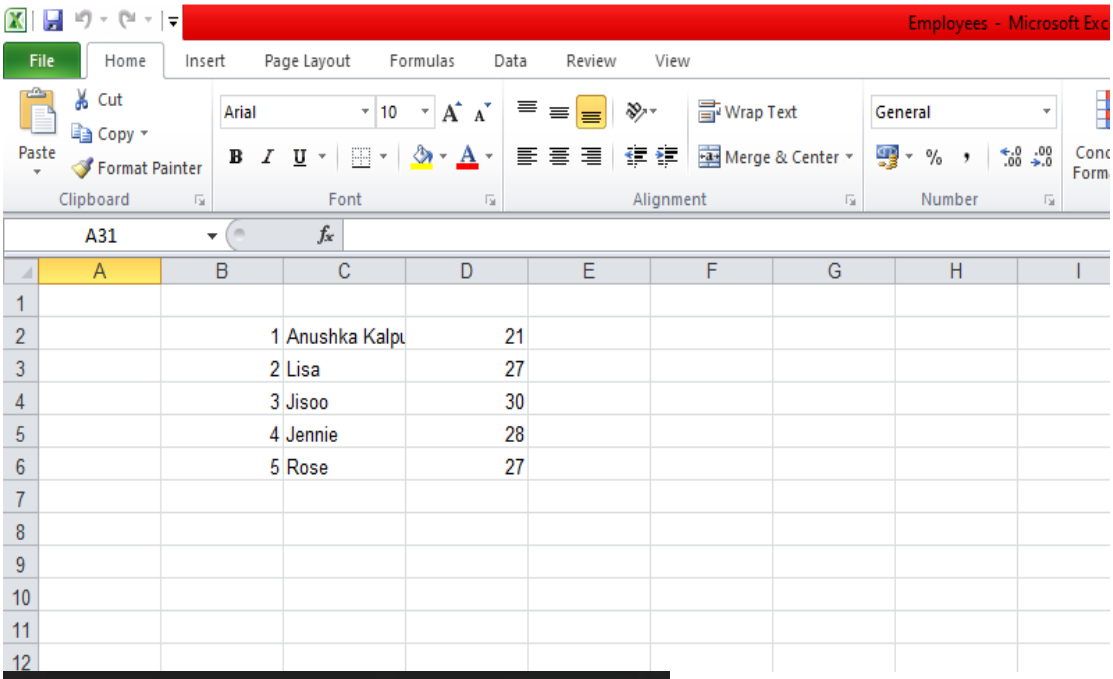


STEP 5

1. Click Run to execute the automation.

2. Open Data.xlsx to check if the table was written correctly.

3. If needed, enhance logging with "Message Box" to show success messages.


Employees - Microsoft Exc
File
Home
Insert
Page Layout
Formulas
Data
Review
View
Cut
Copy
Paste
Format Painter
Clipboard
Arial
10
Font
Wrap Text
Merge & Center
Alignment
General
Number
A31
A B C D E F G H I
1
2 1 Anushka Kalpu 21
3 2 Lisa 27
4 3 Jisoo 30
5 4 Jennie 28
6 5 Rose 27
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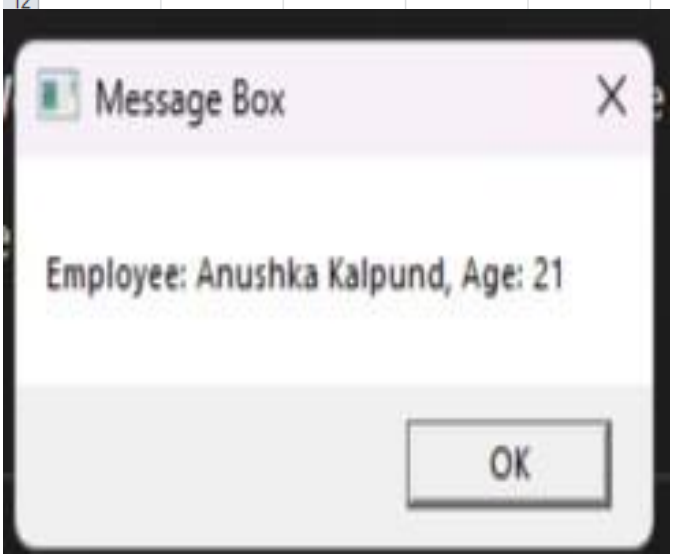
10
11
12


Message Box
Employee: Anushka Kalpund, Age: 21
OK

Experiment 11

Title: Text File: Find and Replace.

Aim: To develop a UiPath workflow that performs **Find and Replace** operations in a text file, thereby automating text manipulation tasks using UiPath Studio..

Objectives:

1. To read content from a text file using UiPath Studio.
2. To identify a specific word or phrase to be searched in the file.
3. To replace the searched word or phrase with a new one.
4. To write the updated content back into the original file or a new file.
5. To understand the application of string manipulation activities in UiPath.

Theory:

In the context of Robotic Process Automation (RPA), text file manipulation is a common and essential task, particularly in business processes where large volumes of logs, reports, or records are handled in plain text format. **UiPath Studio** offers a powerful set of tools to automate file operations such as reading, editing, and writing files, enabling users to save time and reduce errors in repetitive text-processing tasks.

One such text-processing operation is the **Find and Replace** functionality, where a specific word or phrase is searched within the content of a text file and is substituted with another. This operation can be easily automated using a series of string manipulation activities in UiPath, primarily:

- **Read Text File** (from *UiPath.System.Activities*): Reads the entire content of a .txt file and stores it in a string variable.
- **Assign**: Used to perform the actual find and replace operation using the .Replace() method in VB.NET.
- **Write Text File**: Writes the modified string back to a file.

This kind of automation is particularly useful for:

- Mass editing of configuration or log files.
- Generating customized documents (e.g., replacing placeholders like <NAME> with actual data).
- Pre-processing data before input into other systems or software.

In a real-world scenario, businesses can use such automation for dynamic content generation in contracts, replacing outdated links in documents, or updating configuration values in batch.

By building this in UiPath, students and professionals can develop a strong understanding of:

- File I/O operations,
- String manipulation using VB.NET functions,
- Basic workflow design and debugging in UiPath Studio.

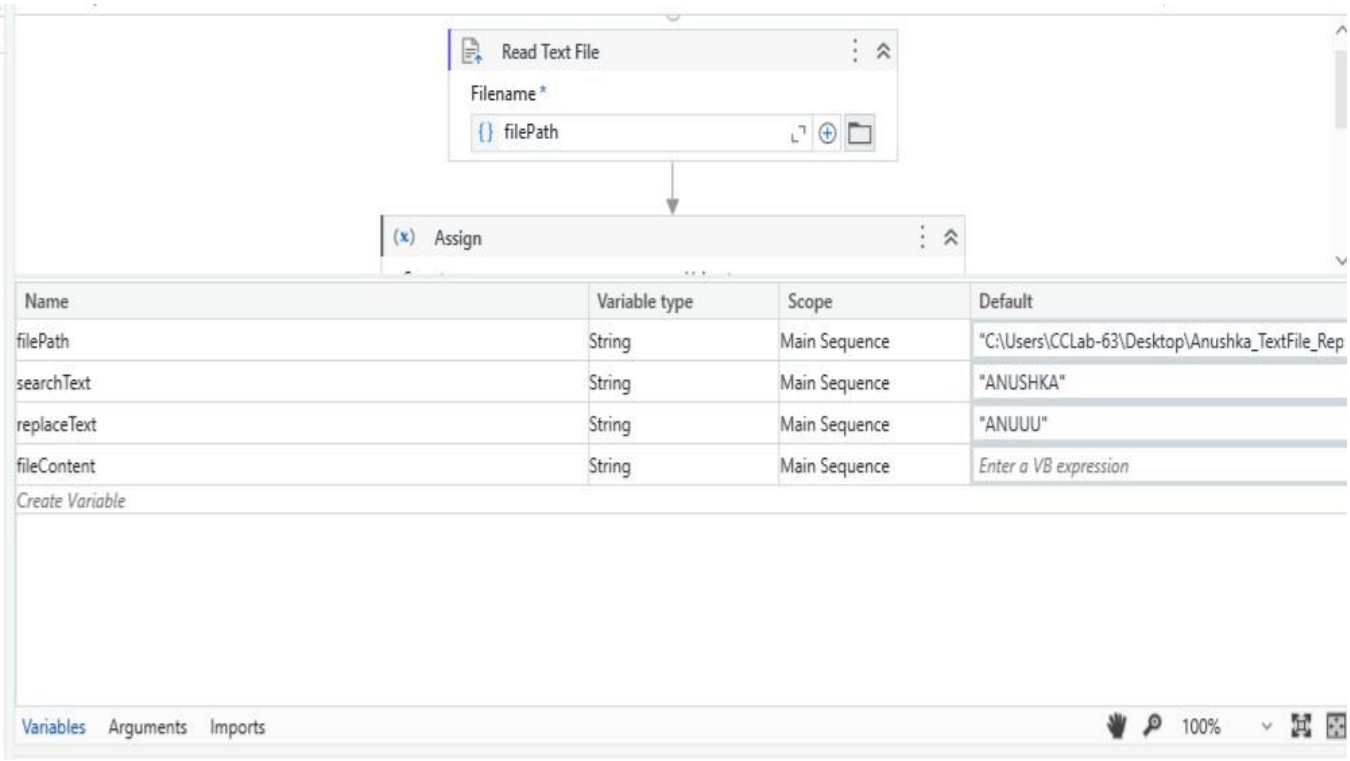
Program Implementation (UiPath Activities Used):

1. **Input Dialog** (optional) – To get the word to find and replace.
2. **Read Text File** – Read content of the source file.
3. **Assign** – Replace old word with the new word using .Replace().
4. **Write Text File** – Save the result to the same or another text file.

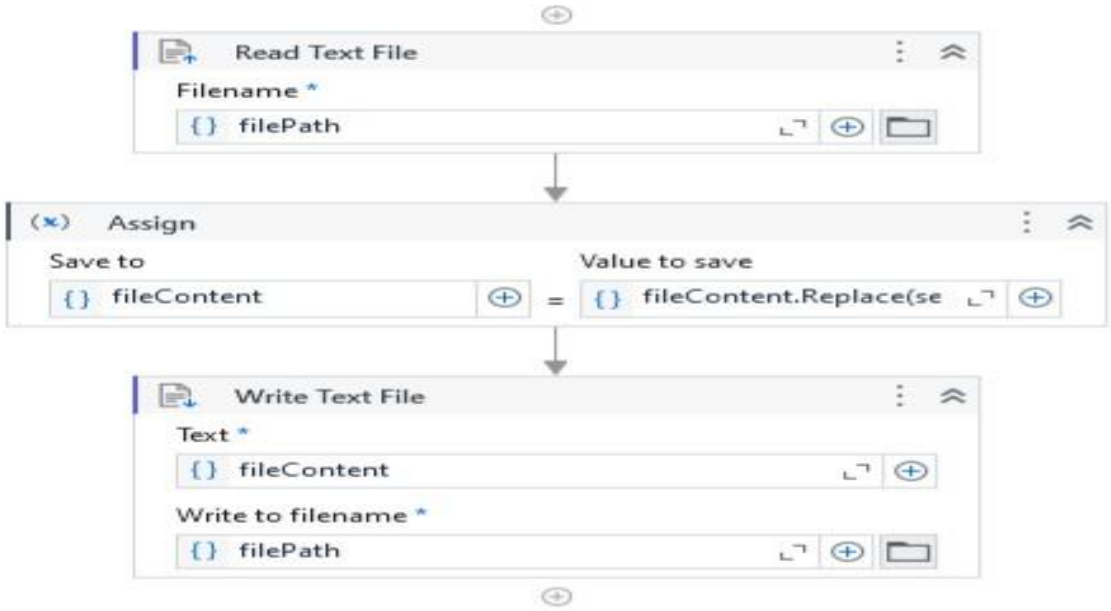
Conclusion :

Through this project, the **Find and Replace** operation on a text file was successfully automated using UiPath Studio. The workflow demonstrates the use of fundamental activities like reading from files, using string functions, and writing outputs, showcasing how UiPath can streamline manual file editing tasks. This automation can be further extended for more advanced use cases like batch processing, dynamic replacement from a database, or integrating with emails and other systems.

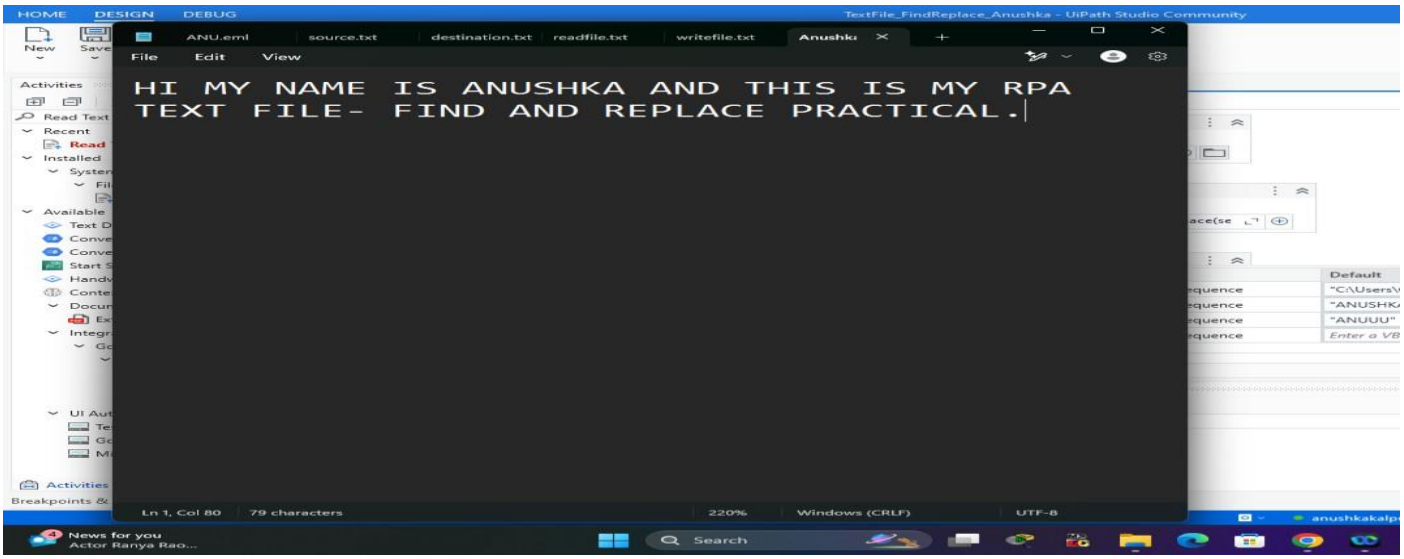
1. Create Varaibles



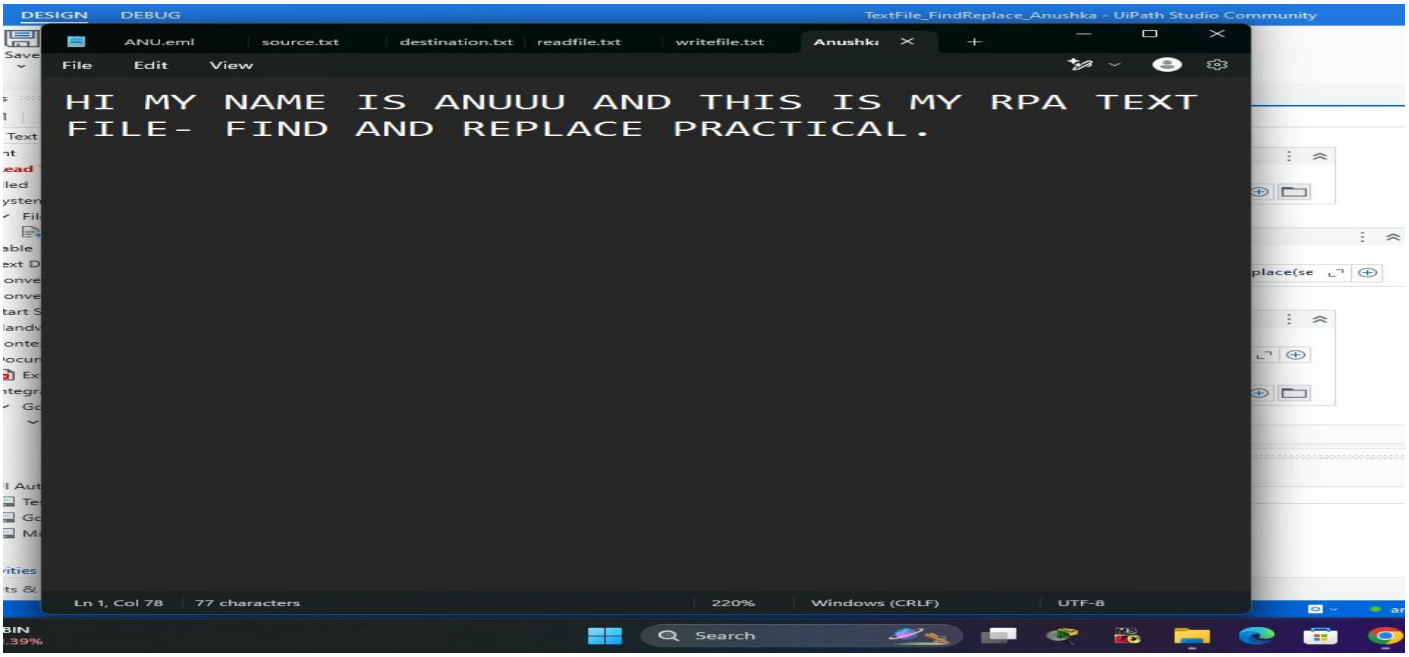
2. Activities



3. Before - Text File



4. After - Text File



Experiment 12

Title: Web Browser Automation.

Aim: To automate web-based tasks such as opening a browser, navigating to a website, extracting or inputting data using **UiPath Studio**, thereby demonstrating web browser automation using Robotic Process Automation (RPA).

Objectives:

1. To understand the use of browser automation activities in UiPath Studio.
2. To automate opening a web browser and navigating to a specified URL.
3. To perform basic web actions such as clicking, typing, and scraping using UiPath activities.
4. To validate the automation by executing the workflow and observing the automated behavior.

Theory:

Introduction to Web Browser Automation

Web browser automation refers to the use of software tools to automatically perform actions within a web browser. This can include navigating to a website, filling out forms, clicking buttons, scraping data, and more. In the context of Robotic Process Automation (RPA), browser automation is one of the most commonly used applications, especially for automating repetitive online tasks like form submissions, data entry, and web scraping.

UiPath Studio is a visual programming tool used to build RPA workflows. It offers a variety of pre-built **browser automation activities** that can interact with UI elements on a webpage. These include:

- Open Browser
- Type Into
- Click
- Get Text
- Extract Data
- Attach Browser

UiPath uses a **UI Automation framework** that identifies UI elements on the screen using selectors (an XML-like structure based on HTML attributes like tag, id, and name). It can work with browsers like **Chrome**, **Firefox**, **Edge**, and **Internet Explorer** after the respective UiPath extensions are installed.

How Web Automation Works in UiPath

The automation typically follows these steps:

1. **Install Browser Extensions**: UiPath requires browser extensions (for Chrome/Edge/Firefox) to interact with the web page.
2. **Open Browser**: This activity launches a specified browser and opens a URL.
3. **Perform Web Actions**: Use Click, Type Into, and Get Text activities to perform operations on the web page.
4. **Data Scraping**: The **Data Scraping Wizard** in UiPath can extract structured data from web pages.
5. **Close or Attach Browser**: Once done, the browser can be closed or reattached for further use.

Use Cases of Web Automation

- Auto-login to web portals
- Submitting online forms
- Extracting product prices from e-commerce sites
- Reading news headlines from websites
- Automating email web clients (e.g., Gmail)

Key Benefits

- **Time-saving**: Speeds up repetitive tasks
- **Accuracy**: Reduces human errors
- **Scalability**: Can run on multiple machines
- **Integration**: Easily integrates with other desktop and web apps

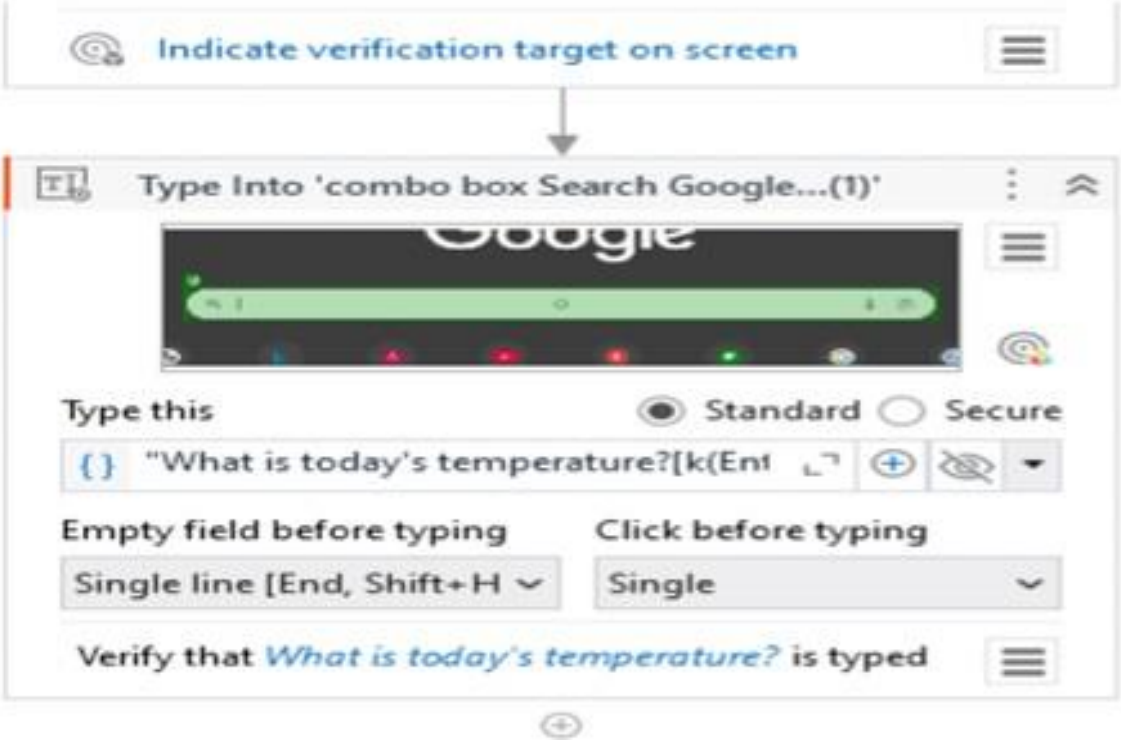
Important Considerations

- Ensure browser extensions are enabled.
- Selectors should be made dynamic for reliability.
- Use delays and retries for loading-sensitive elements.
- Avoid using hard-coded values when possible.

Conclusion :

Web browser automation using UiPath Studio simplifies repetitive web tasks and increases efficiency by mimicking human actions in the browser. With drag-and-drop activities and intelligent UI element recognition, even non-programmers can build powerful automation workflows. This capability plays a vital role in modern industries for tasks like data scraping, form submission, auto-login, and web testing.

Chrome New Tab
Browser URL *
"chrome://newtab/"
Do
Click 'combo box Search Google...'
Click type
Mouse button


Indicate verification target on screen
Type Into 'combo box Search Google...(1)'
Type this
Standard
Secure
"What is today's temperature?[k(Ent
Empty field before typing
Click before typing
Single line [End, Shift+H
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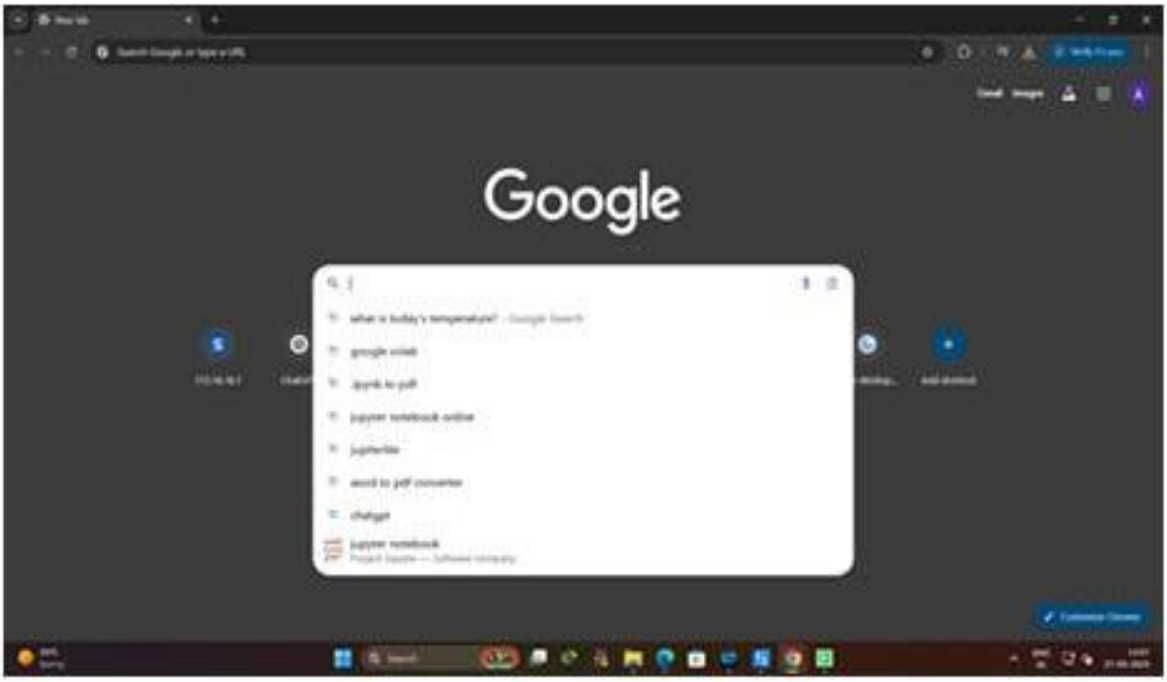
Verify that What is today's temperature? is typed


Google
what is today's temperature? - Google Search
google colab
ipynb to pdf
jupyter notebook online
jupyterlite
word to pdf converter
chatgpt
jupyter notebook

Google
What is today's temperature?
37
Weather
Thursday
Sunday
Temperature
Precipitation
Wind
Thu
Fri
Sat
Sun
Mon
Tue
Wed
Thu
People also ask
Is Pune hot or cold?
What is the climate in Indore today?